Depth 0

Geometric shape representations in human adults have syntactic structure

Fondation Fyssen

Funded by
the European Union

European Research Council
Established by the European Commission

Inserm

La science pour la santé

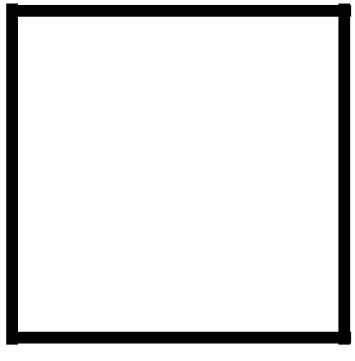
From science to health

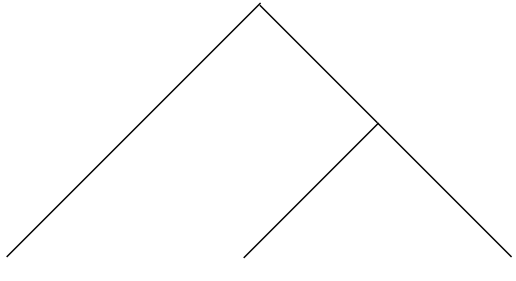
NeuroSpin

Barbu Revencu and Stanislas Dehaene

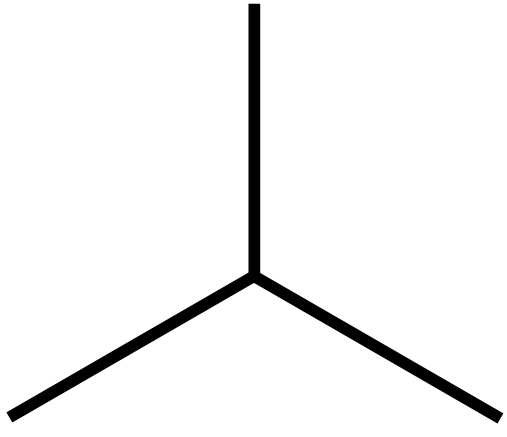
NeuroSpin Cognitive Neuroimaging Unit | CEA | INSERM | Université Paris-Saclay

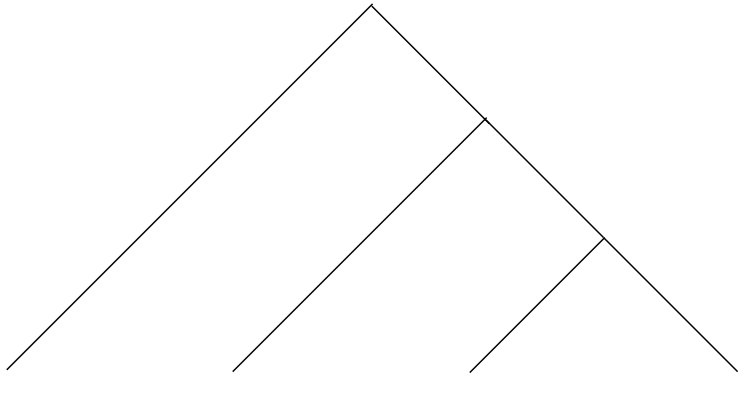
trace
segment





repeat 4 times | **trace** segment | **turn** 90°





repeat 3 times | **execute** & **return** | **trace** segment | **turn** 120°

cea

PREREGISTERED

Targets

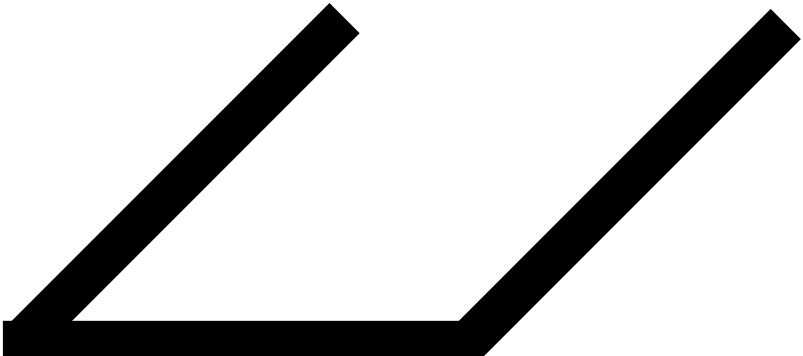
Structure

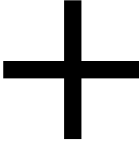
Matches

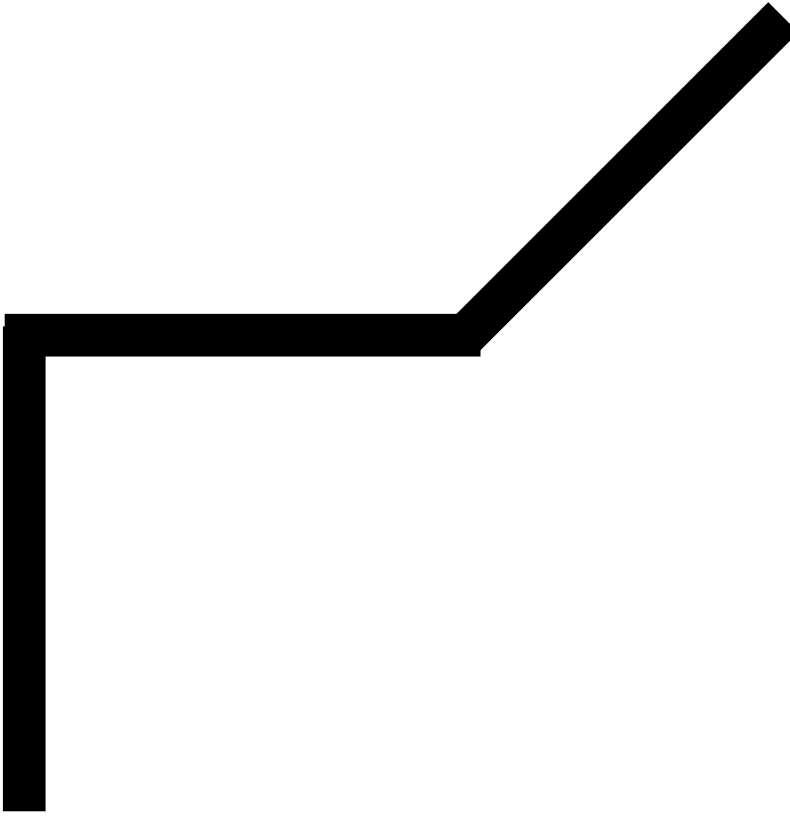
Deviants

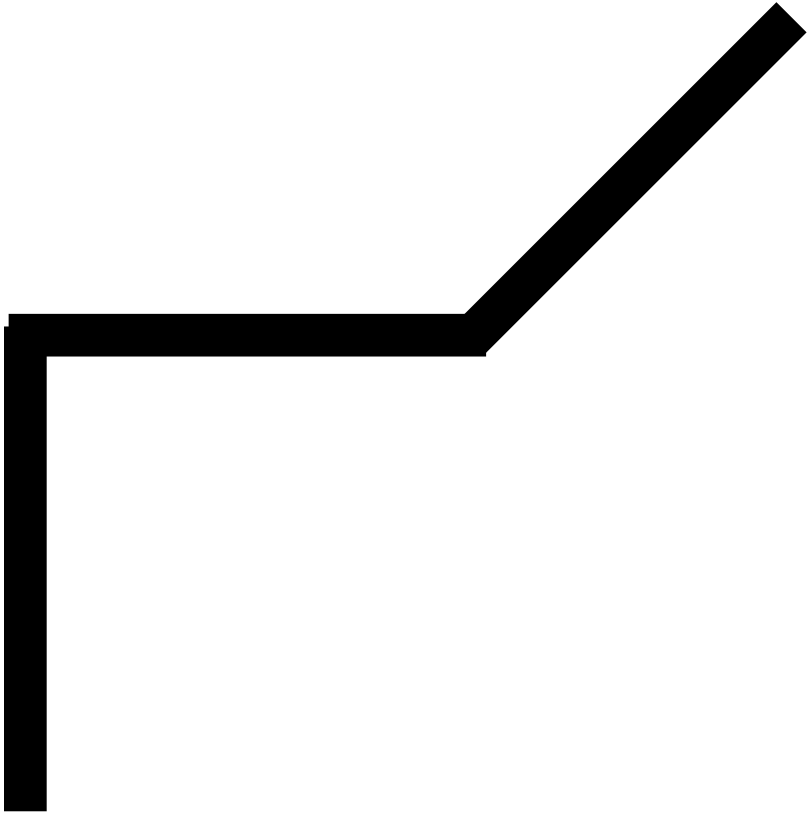
Shape

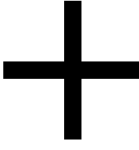
Samples

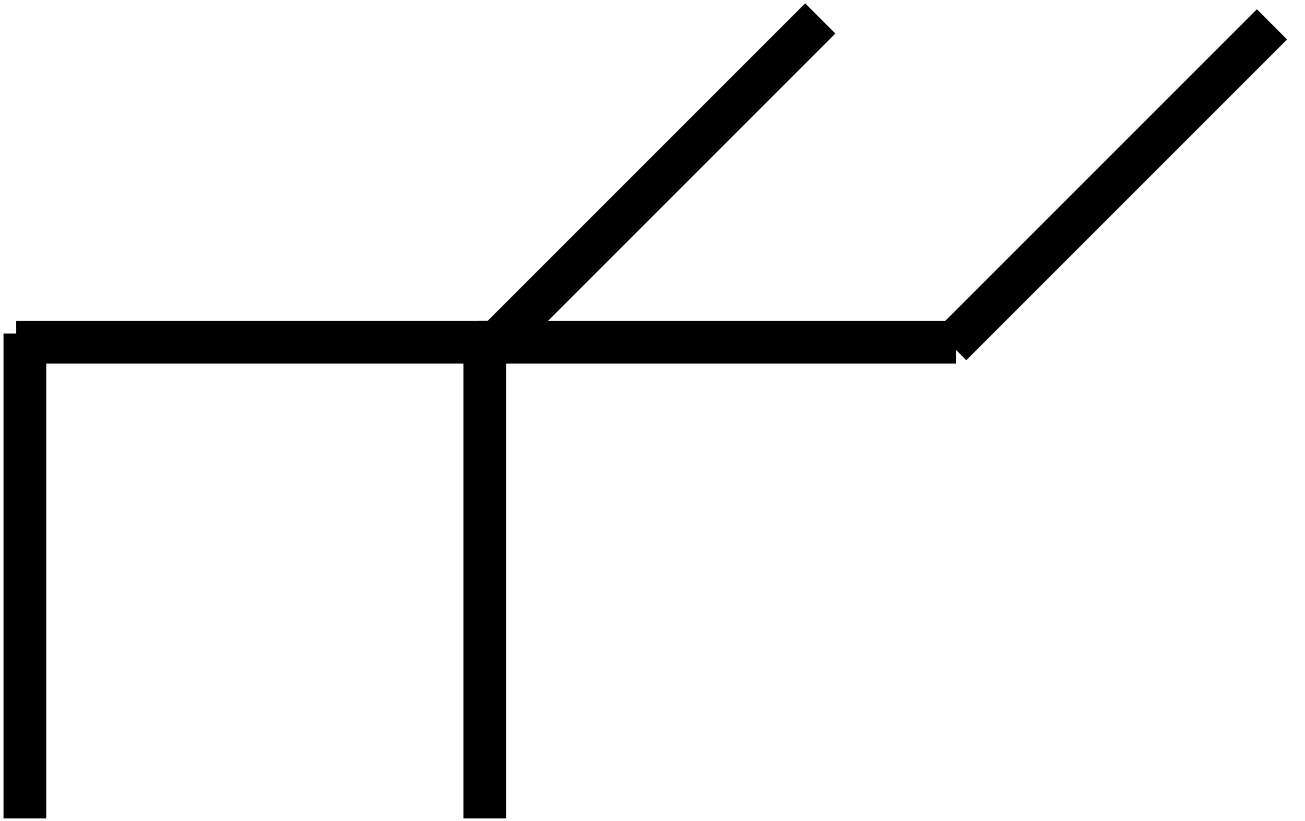


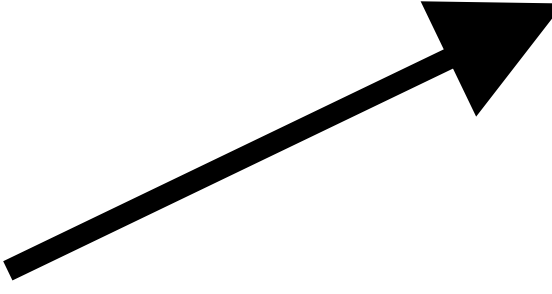


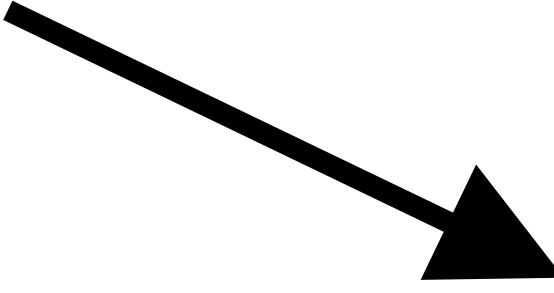


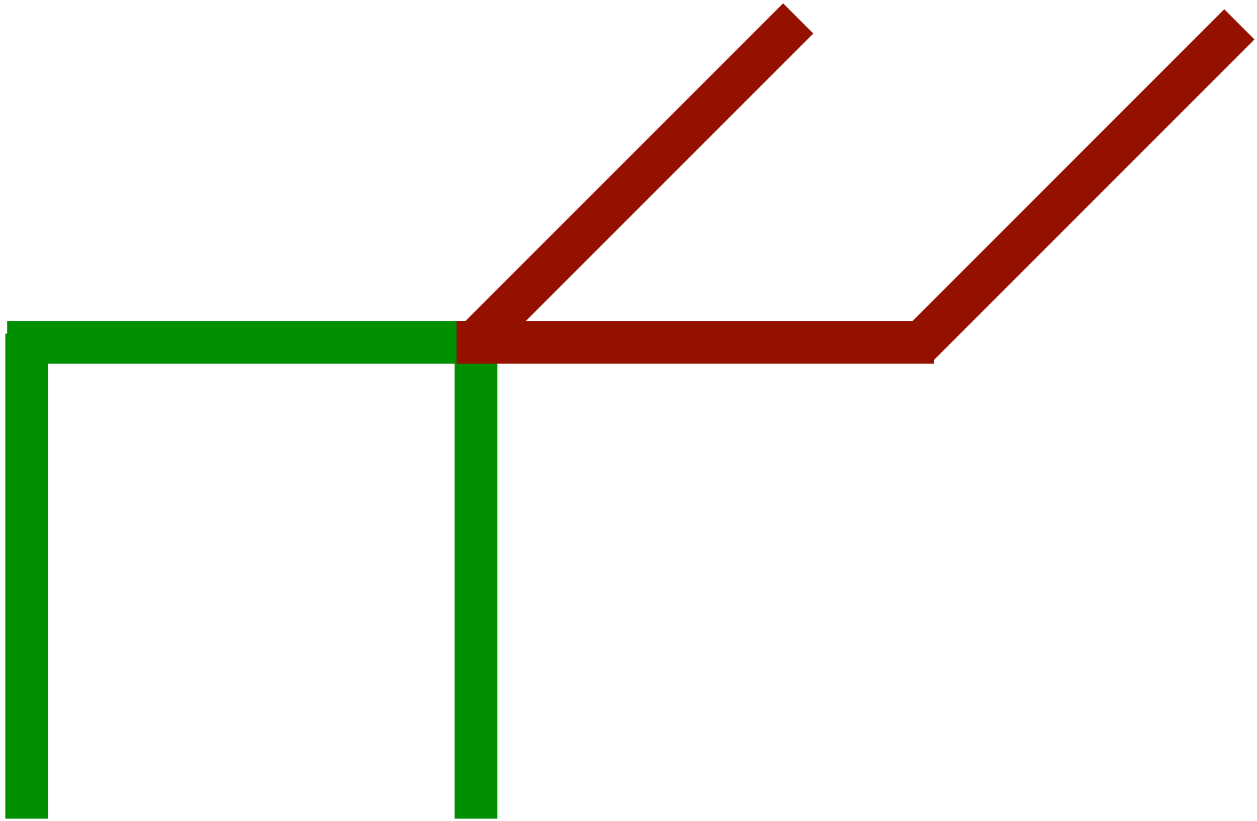


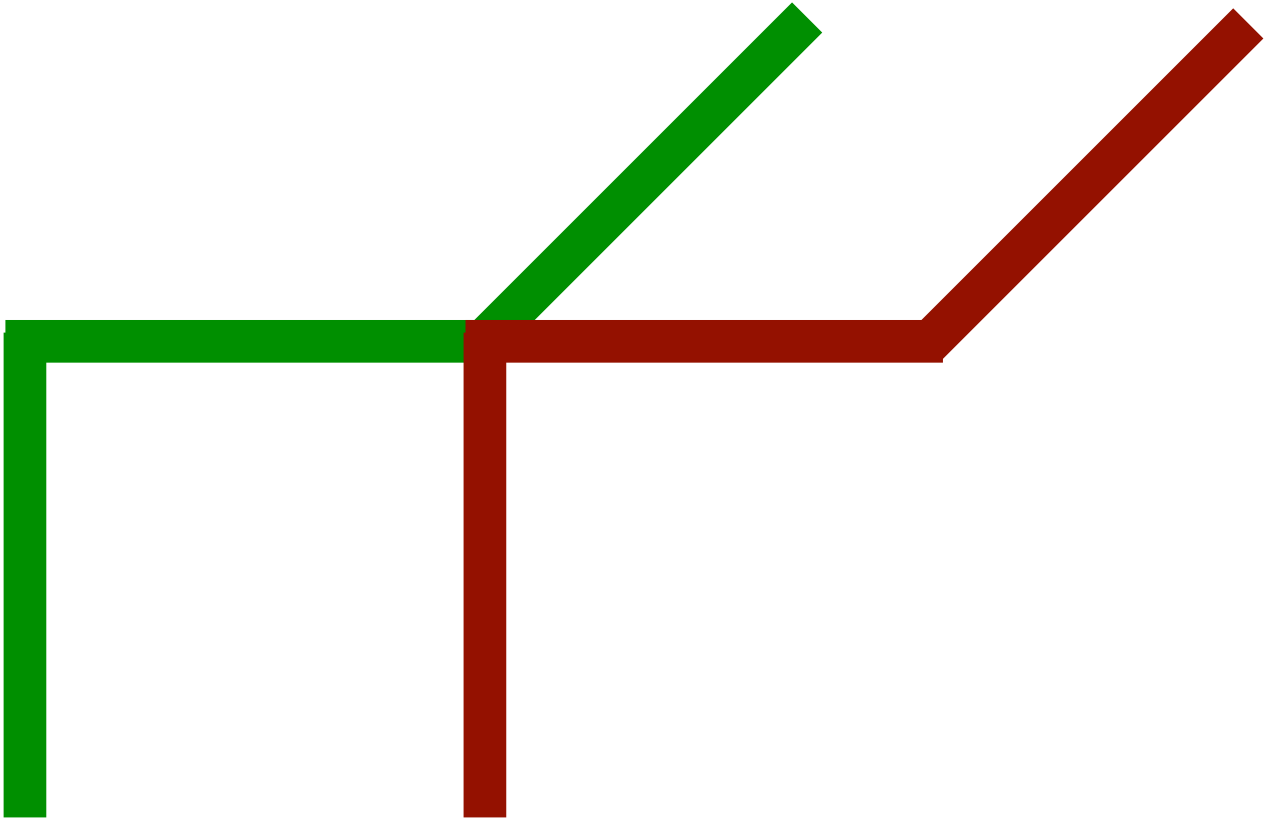








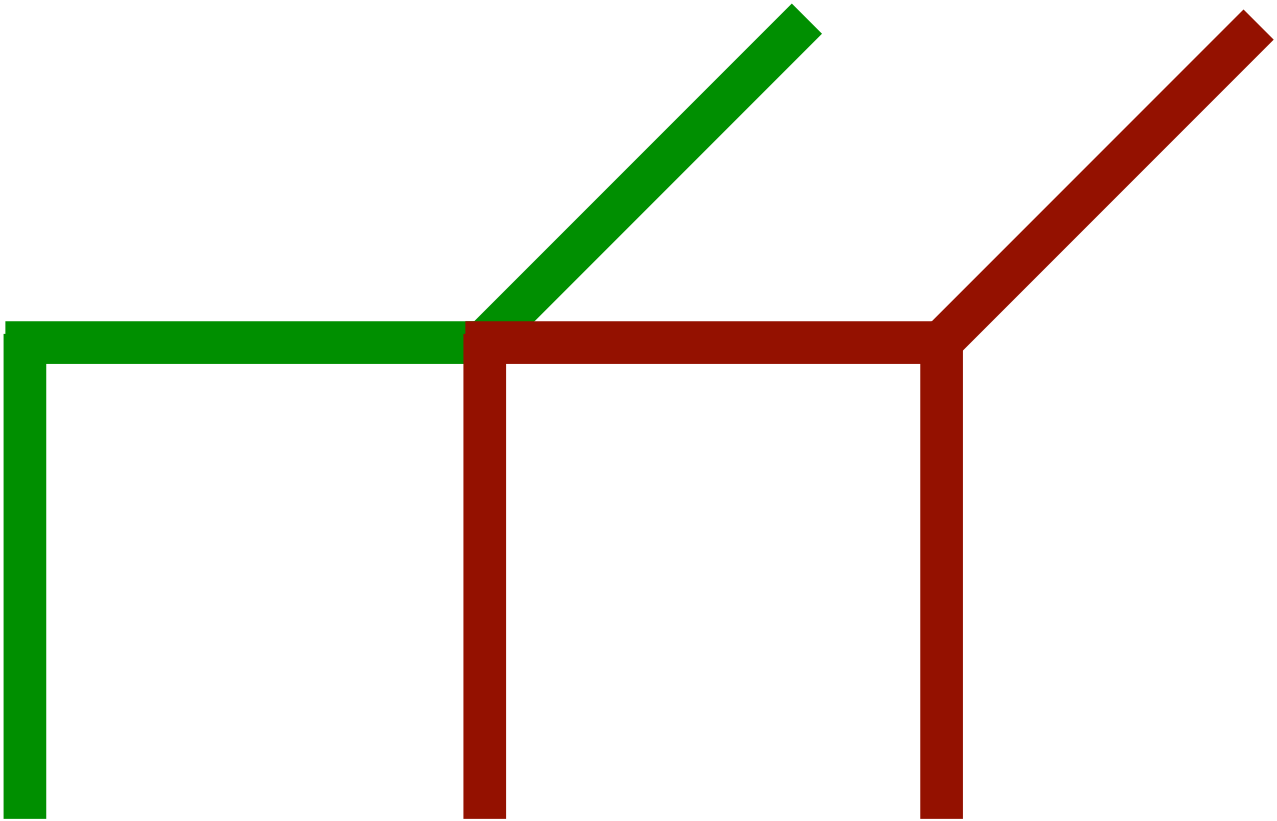


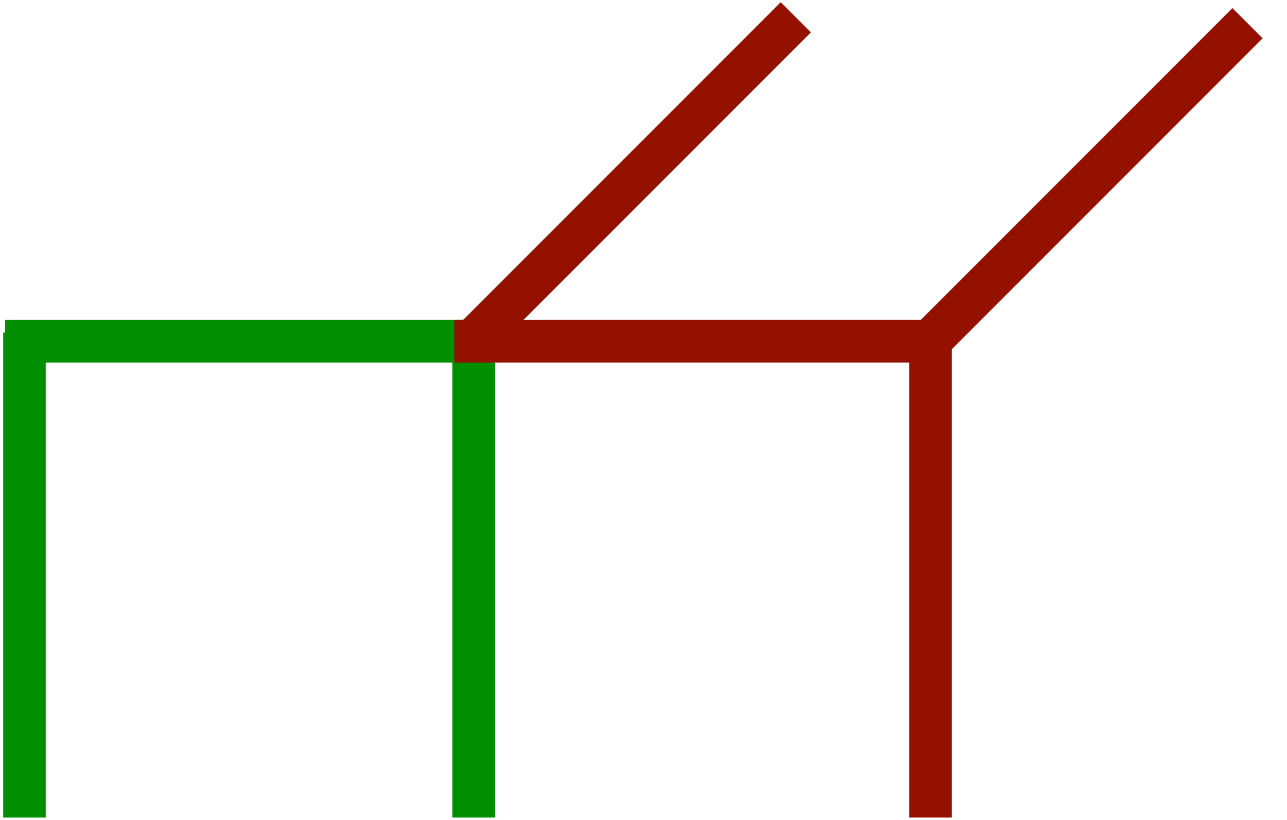


Congruent

Incongruent

Different structured representations can be induced for the same shape





Modified congruent

Modified incongruent

Hypothesis: Geometric shapes are encoded in a dedicated, syntactically structured language of thought

Subparts are easier to recognize when they belong to the same subtree

Shapes are easier to reconfigure when they are split higher up in the tree

Targets

Structure

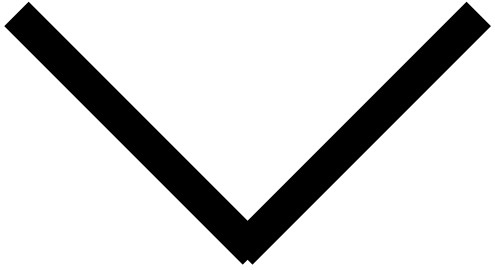
Common part

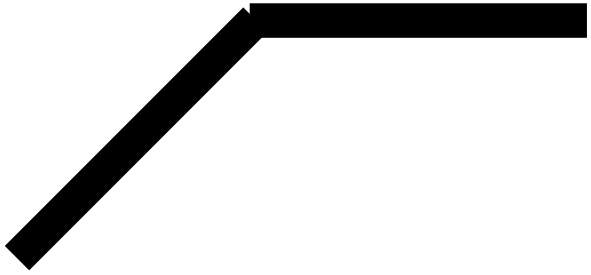
Matches

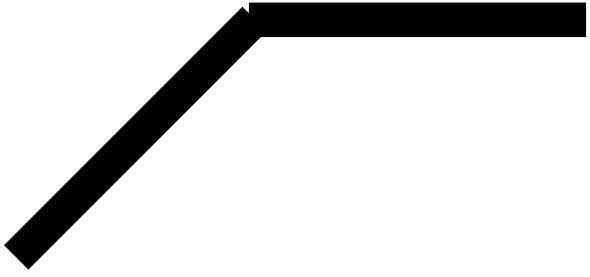
Deviants

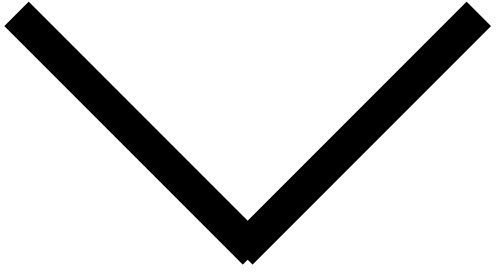
Shape

Samples







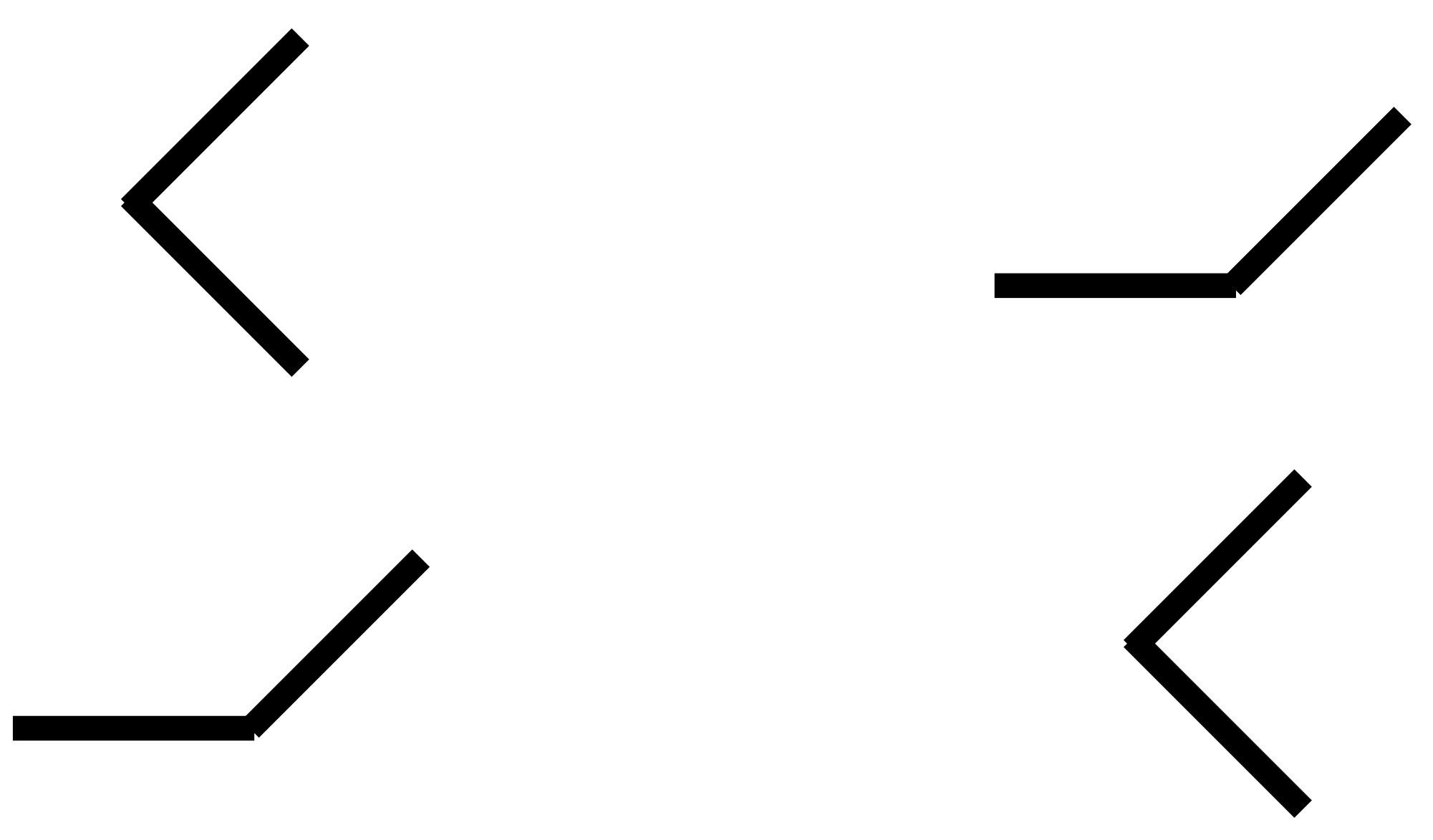


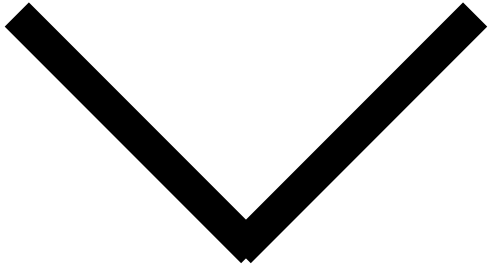
Same subtree

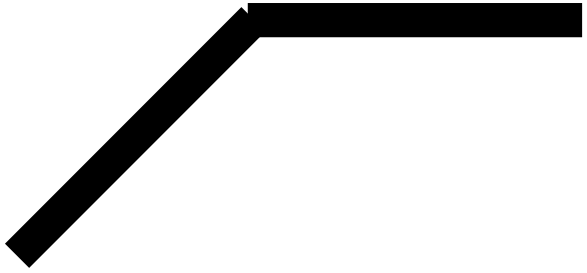
Different subtrees

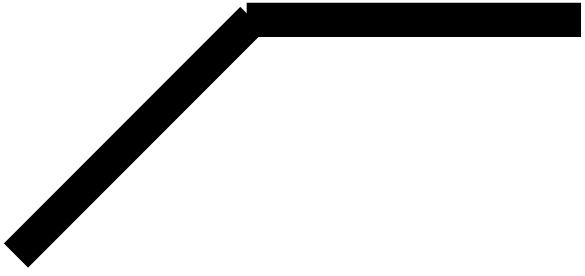
Rotated same

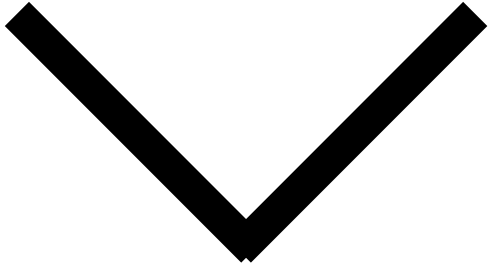
Rotated different

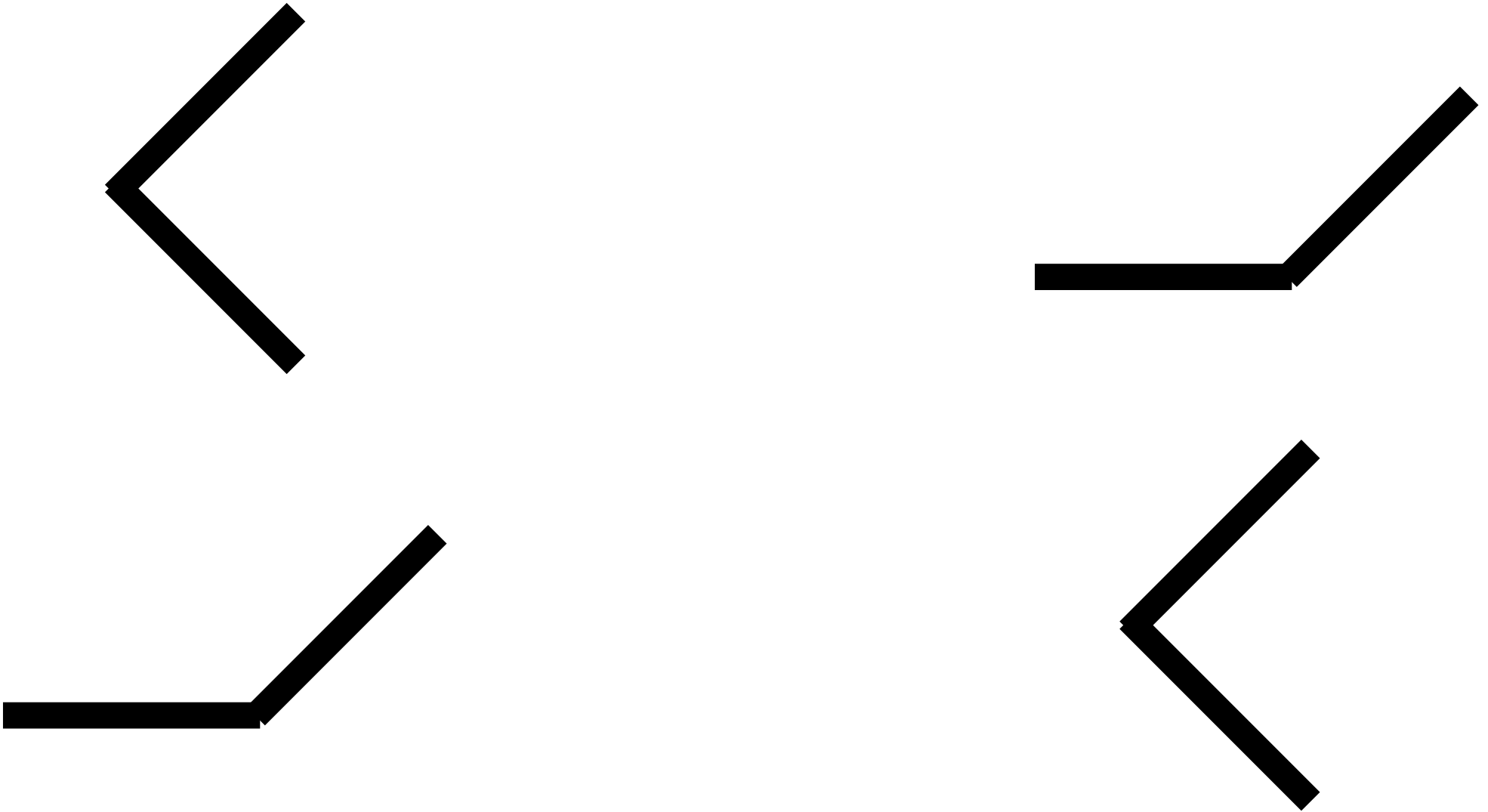


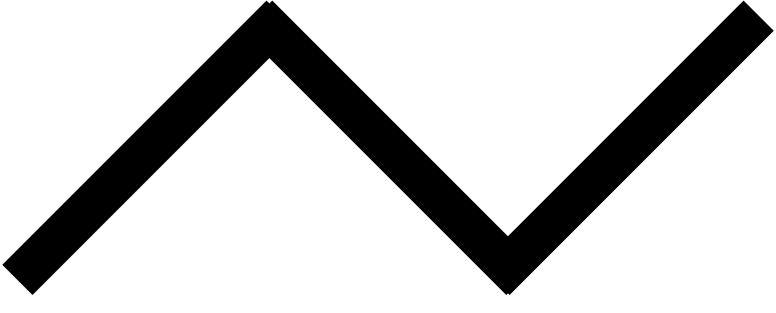












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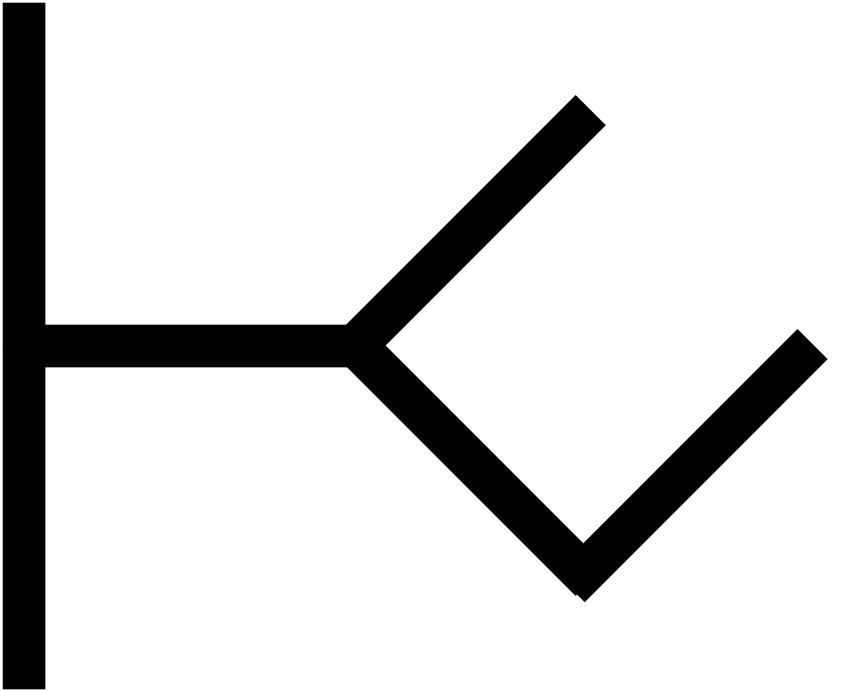
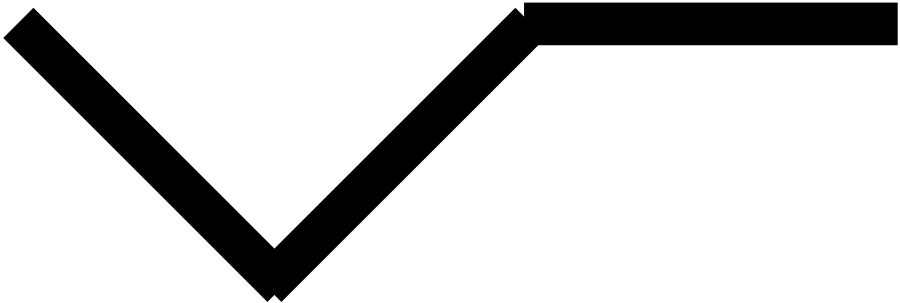
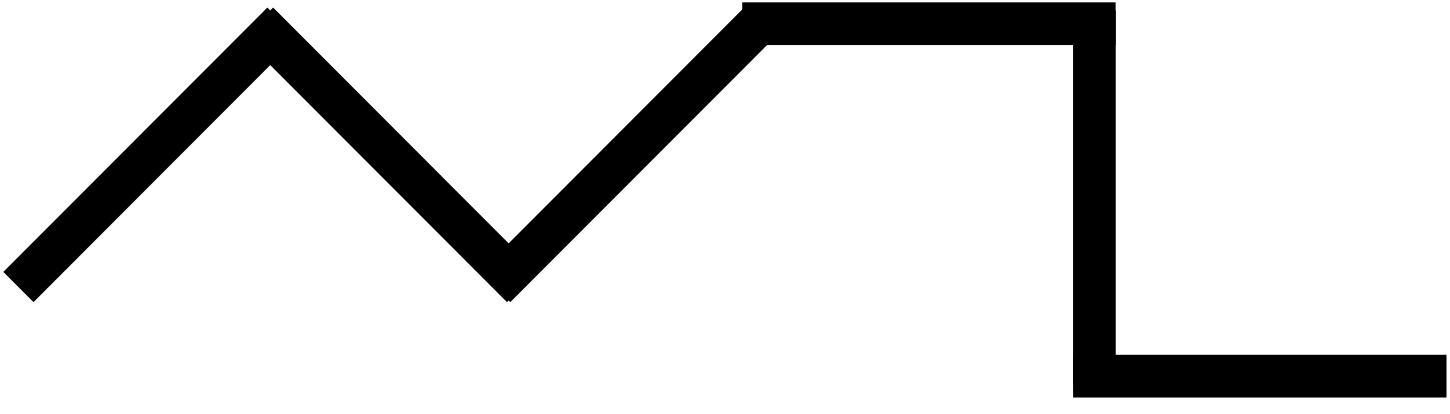
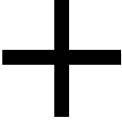
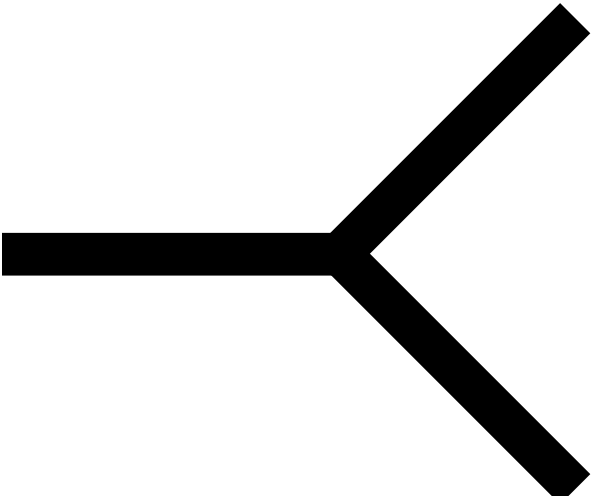
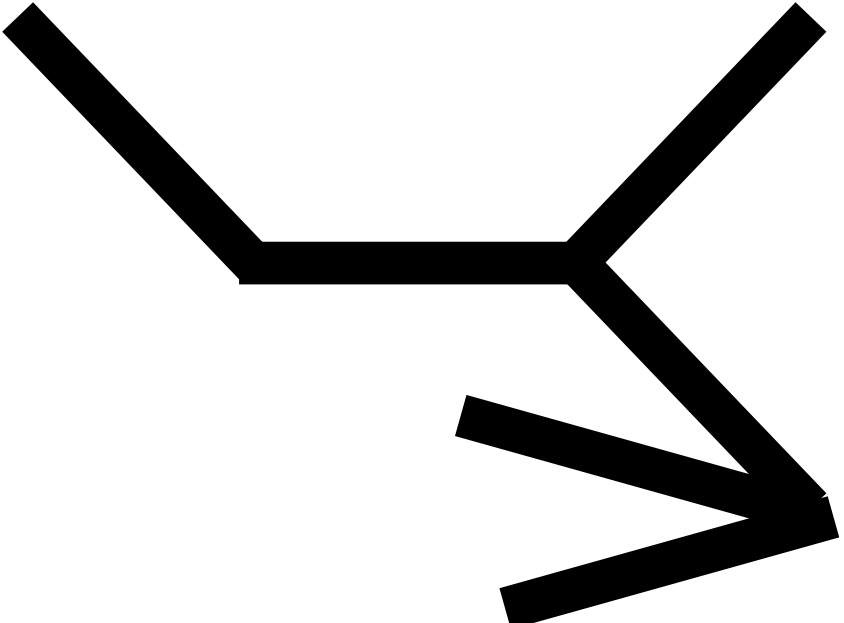


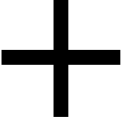


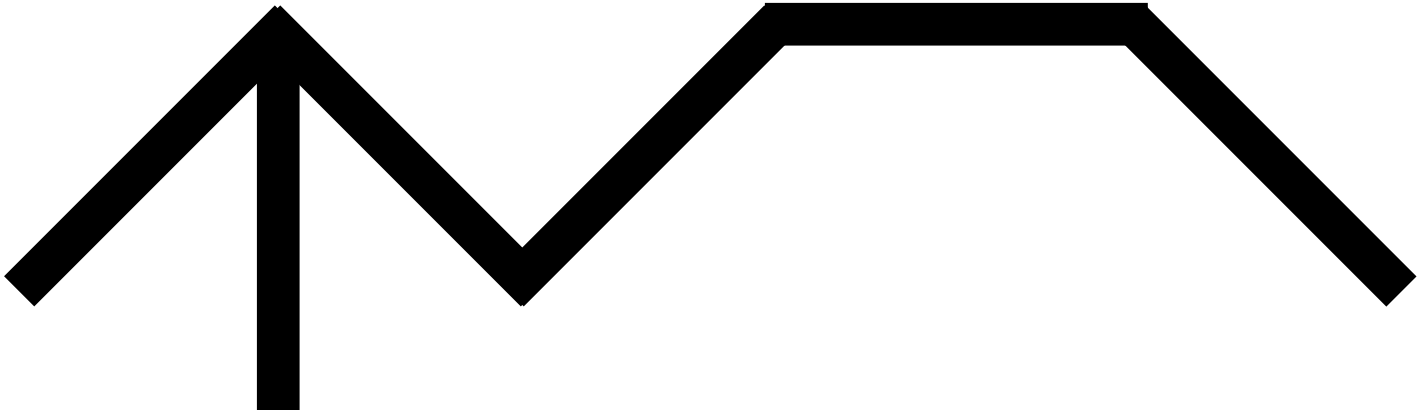


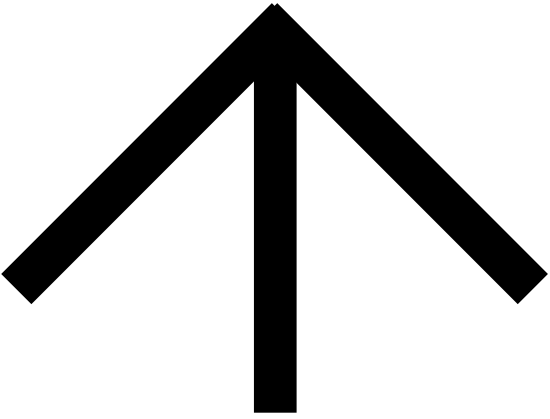


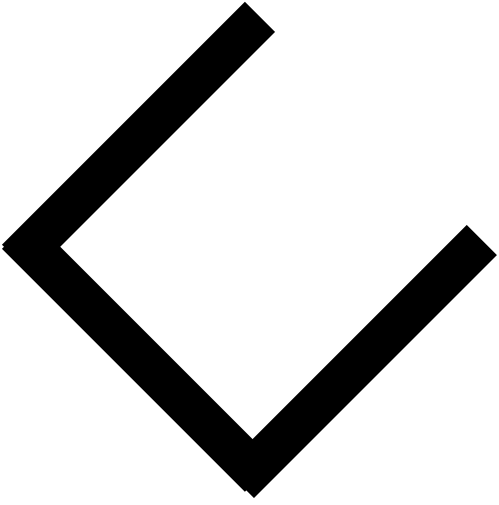


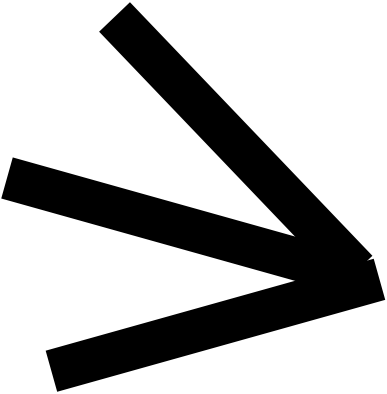


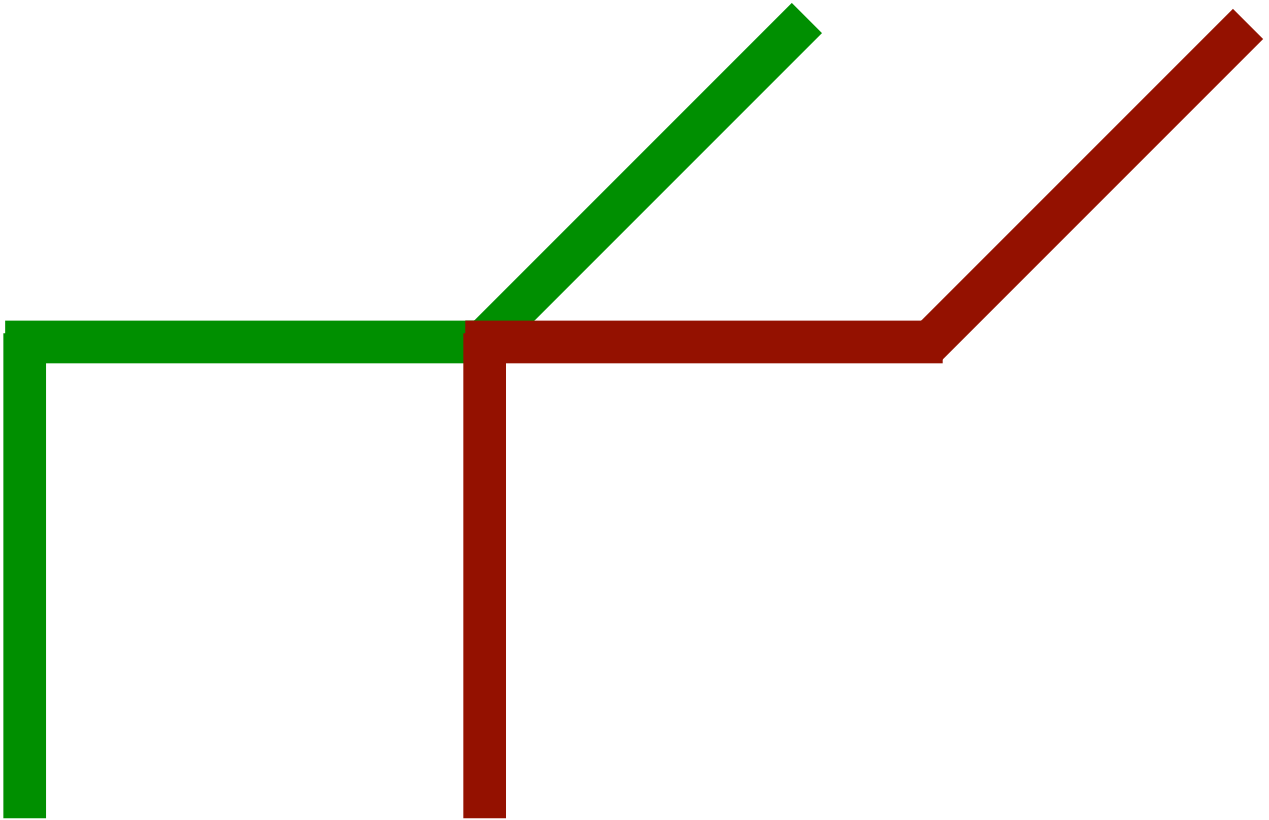


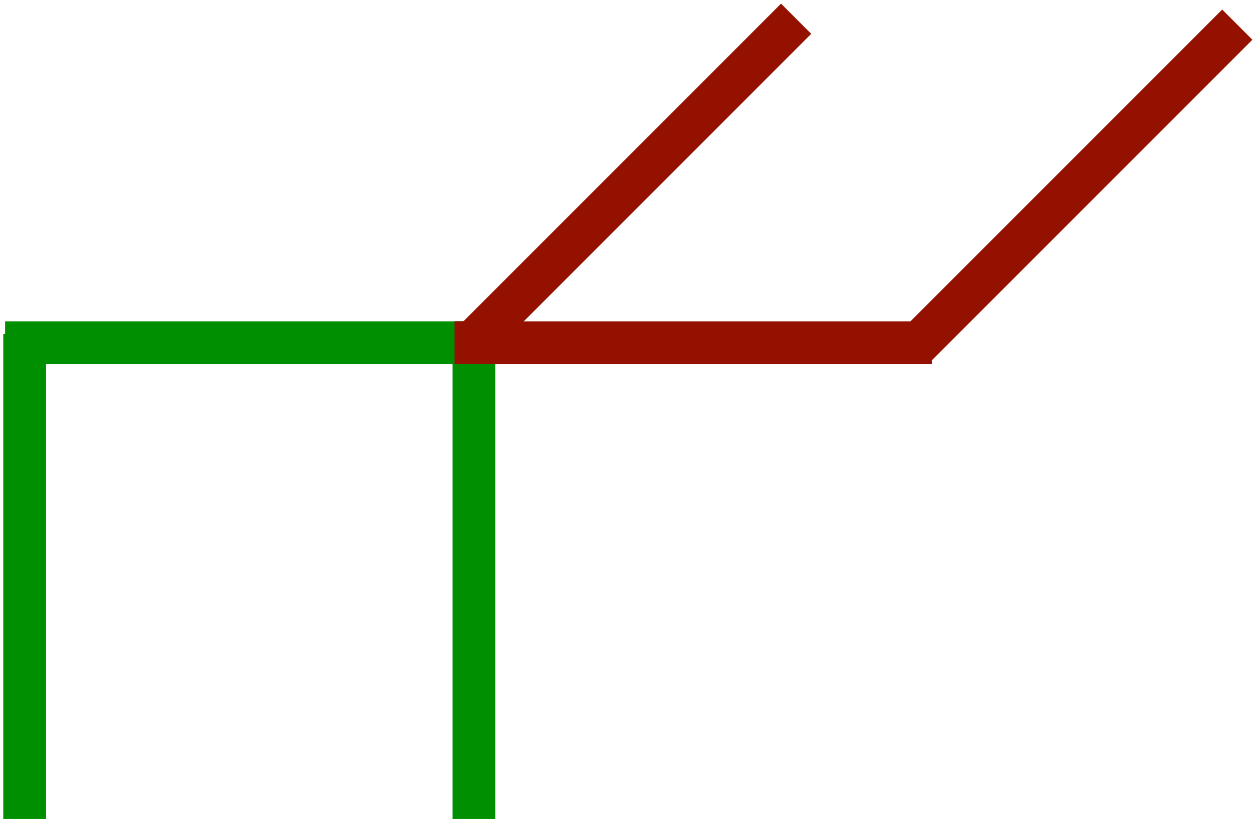


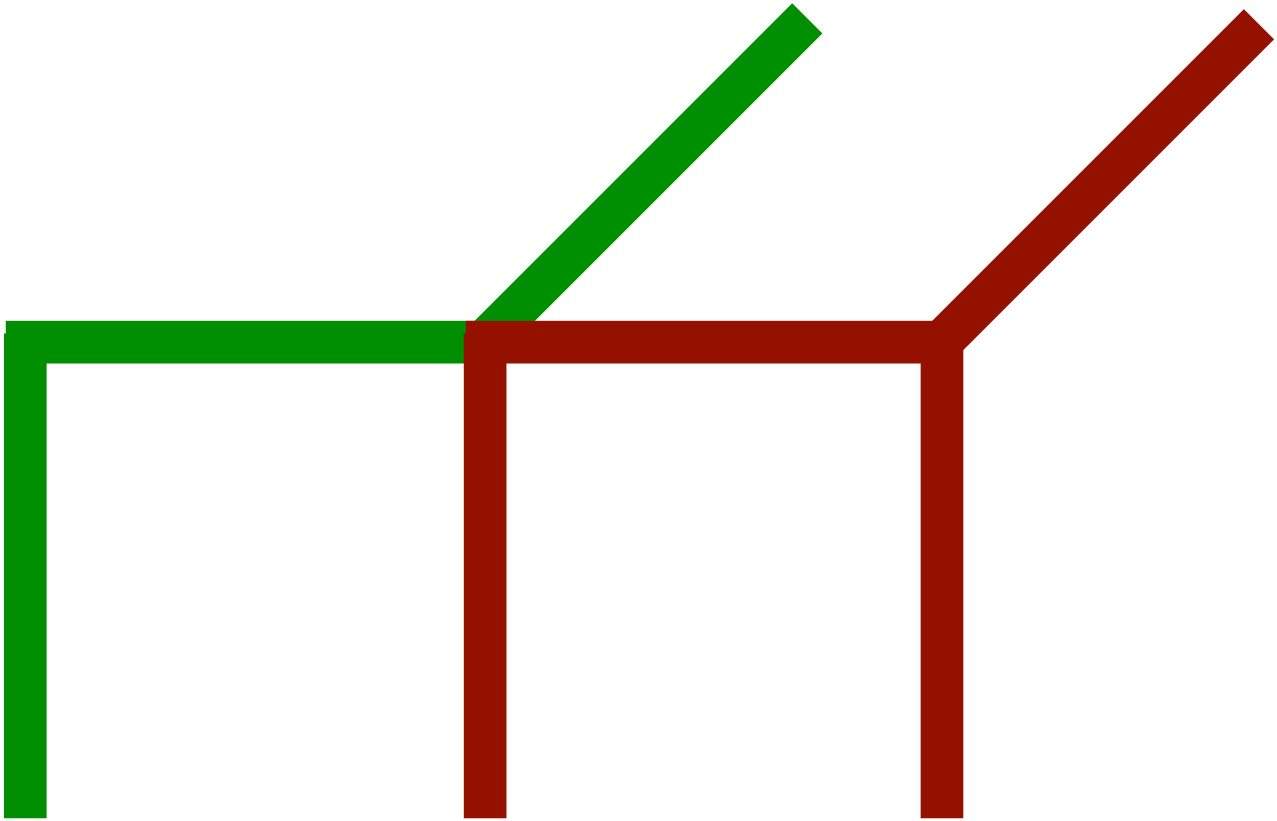


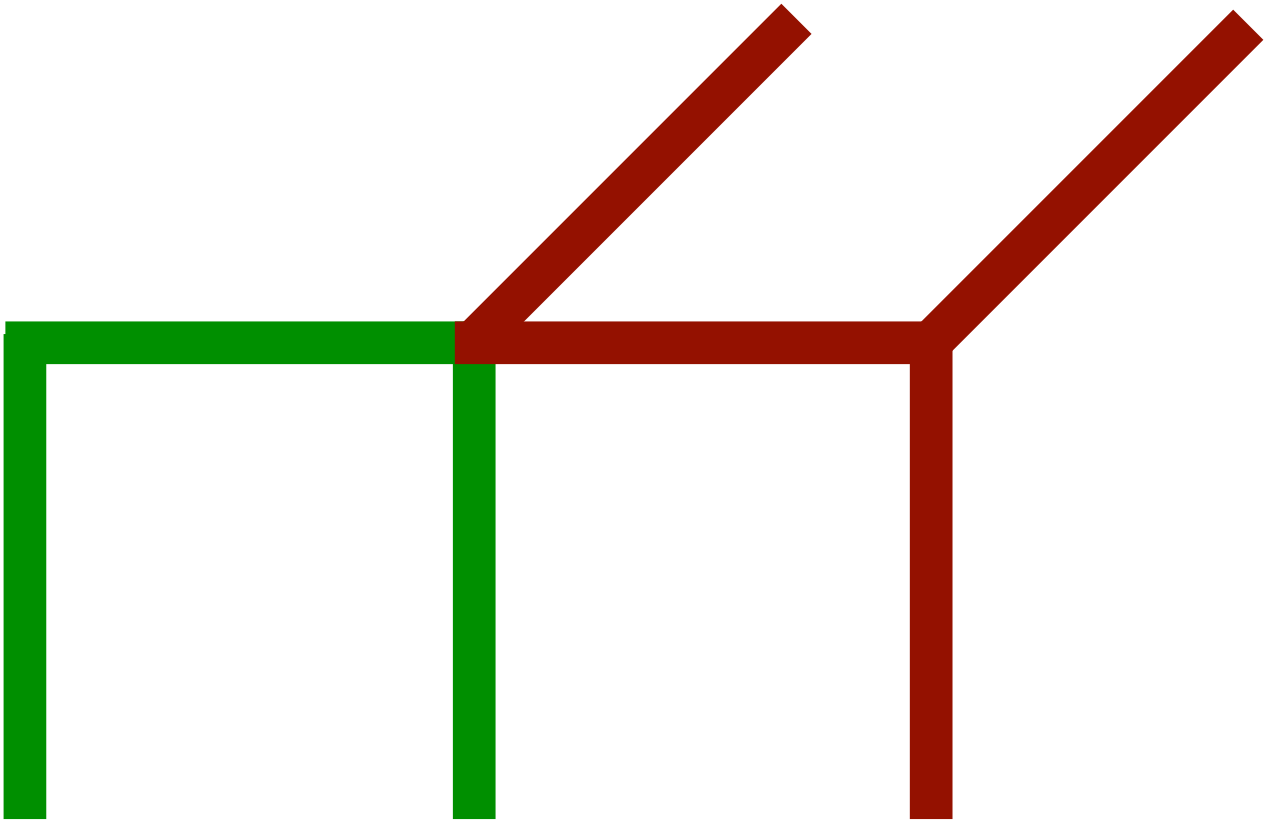












Samples

Shape

Structure

Targets

Matches

Deviants

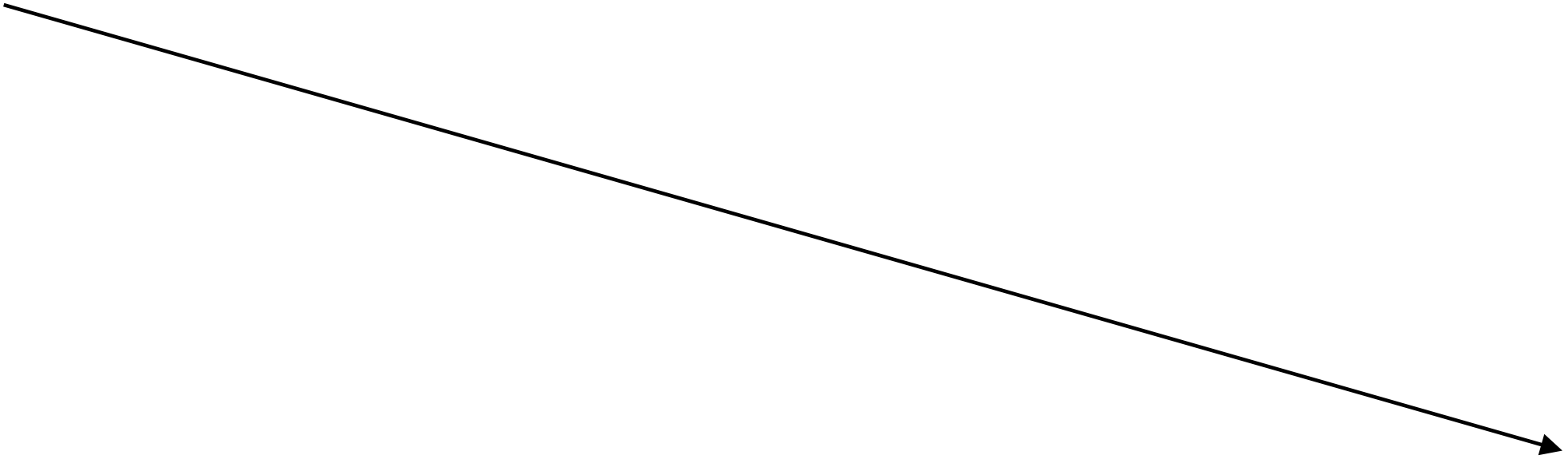
Tree preserved

Tree broken

Tree preserved

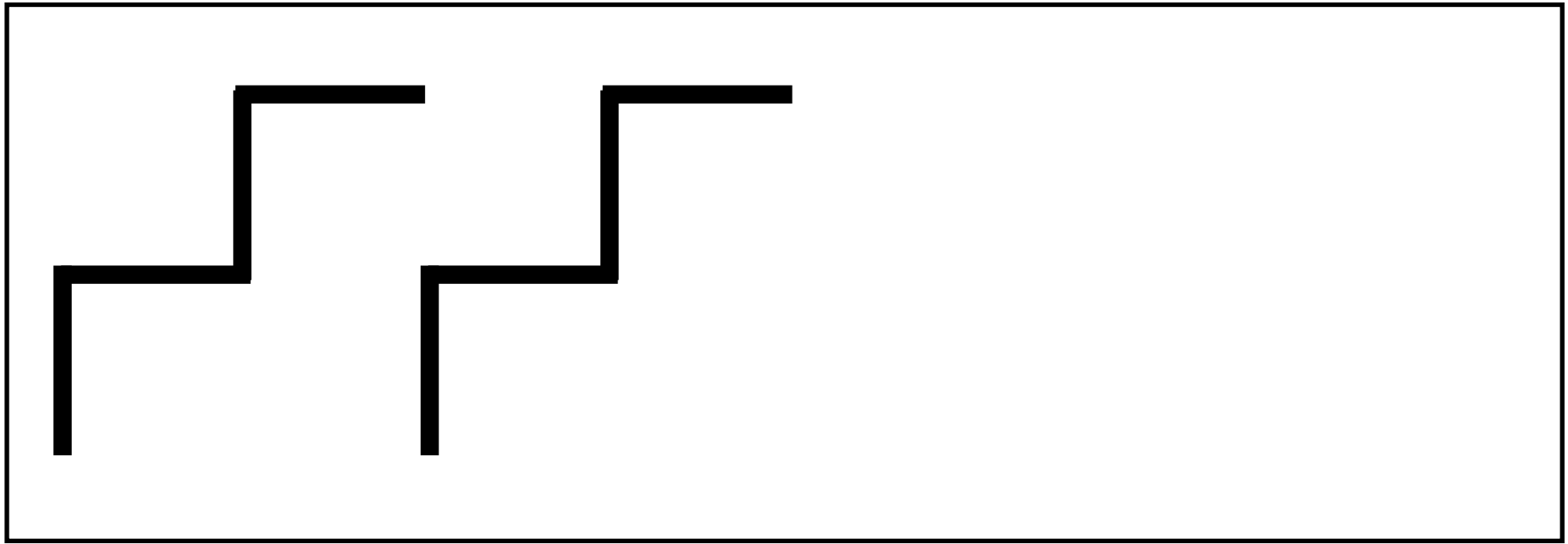
Tree broken

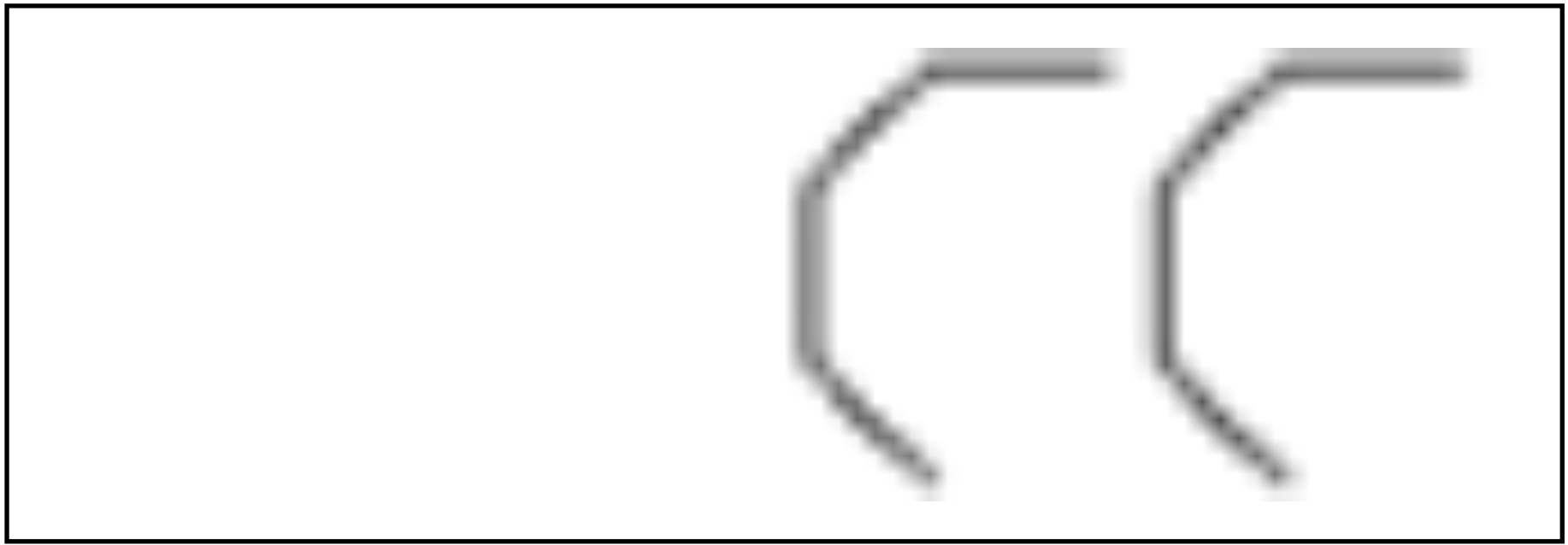
Depth 1



Depth 3

Depth 2





Experiment 1: Structural Ambiguity

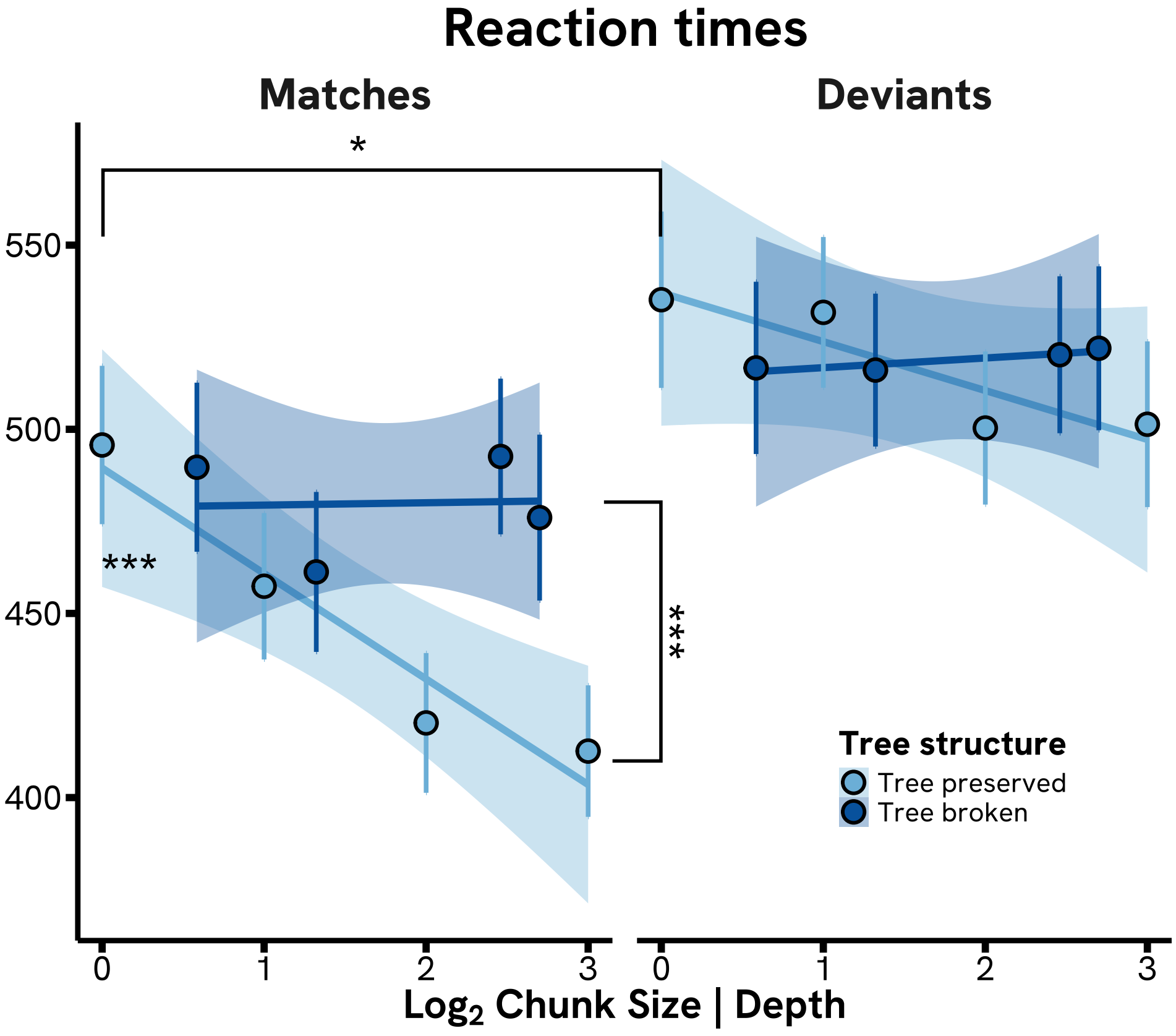
Experiment 2: Subtree Facilitation

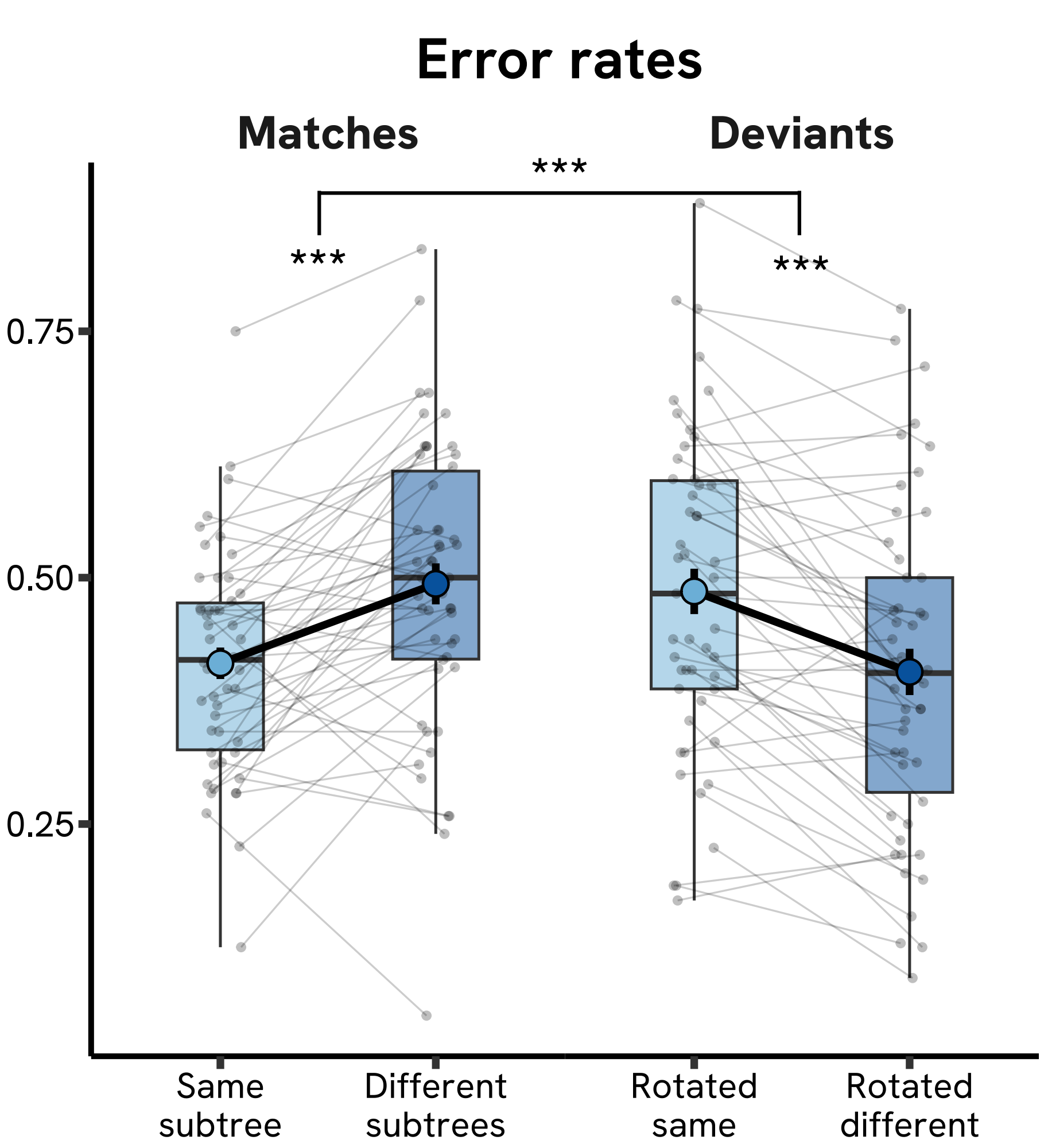
Experiment 3: Movement Depth

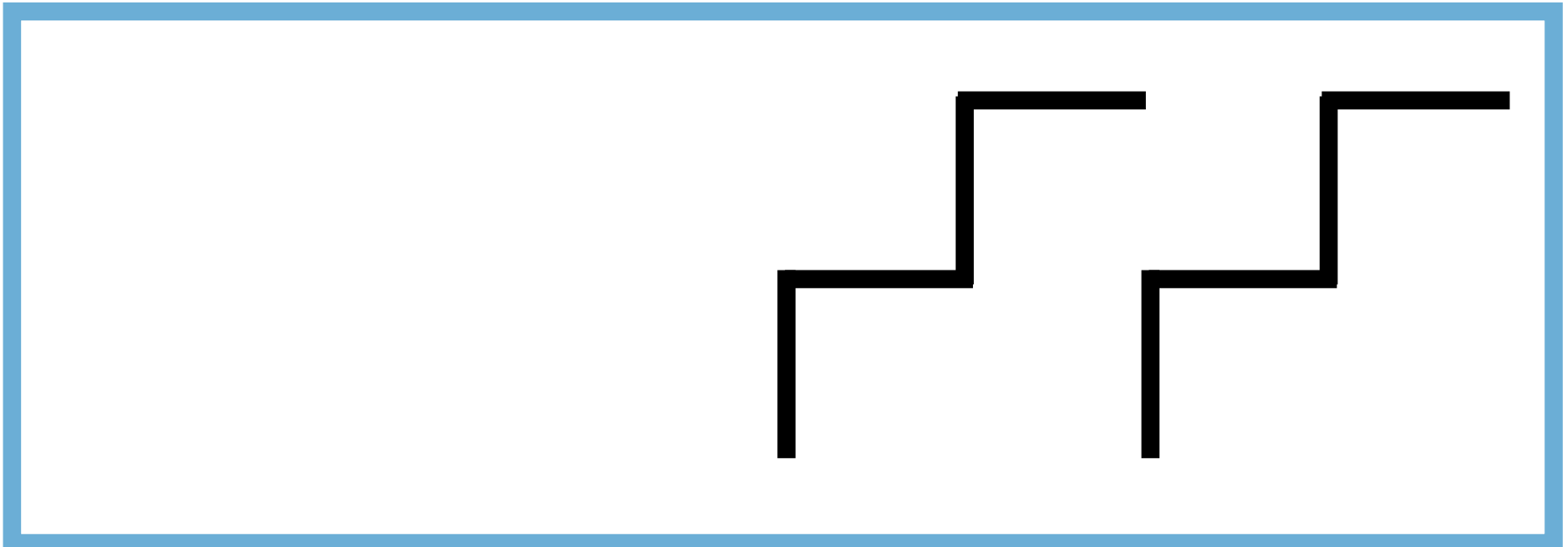
50 subjects | 256 trials

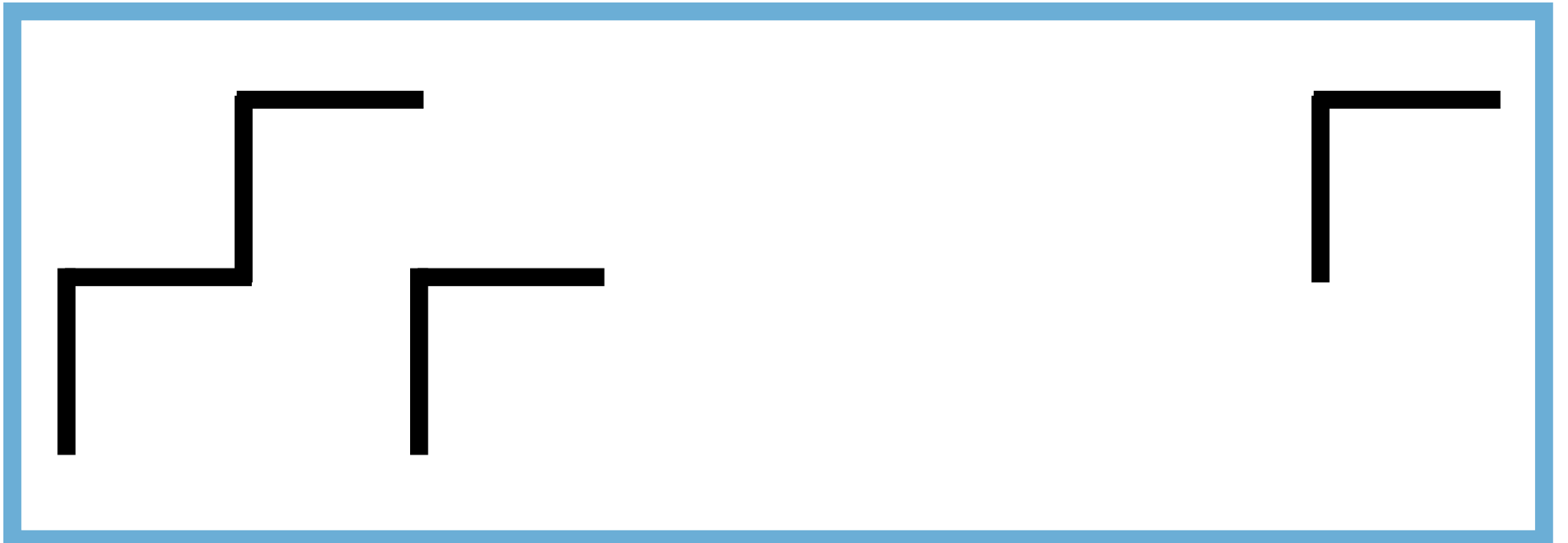
50 subjects | 128 trials

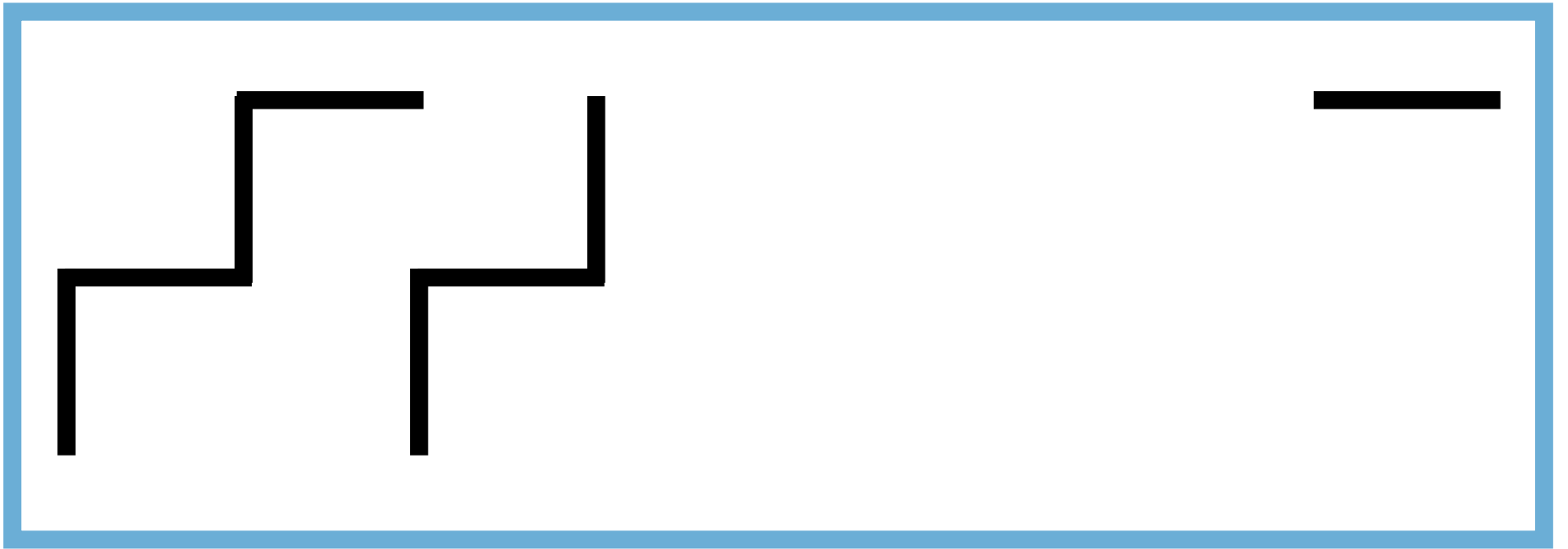
36 subjects | 256 trials

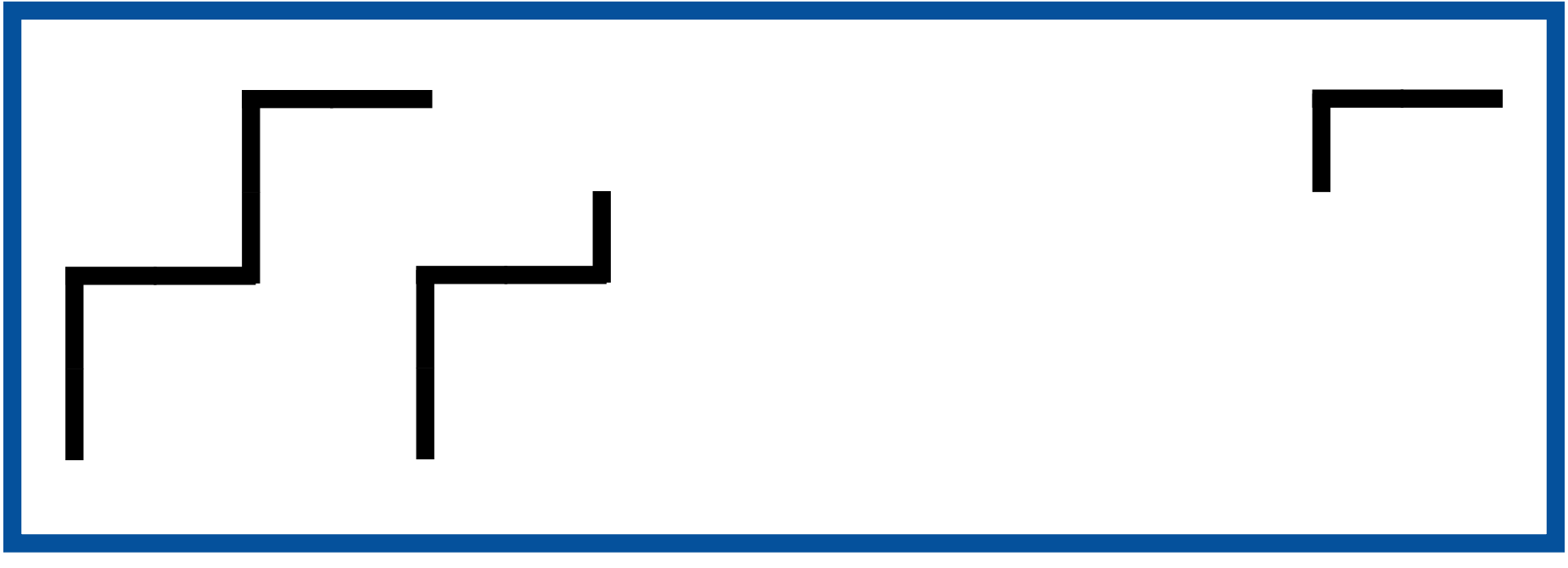

Reaction times
Matches
Deviants
*
***
***
550
500
450
400
0
1
2
3
0
1
2
3
Log2 Chunk Size | Depth
Tree structure
Tree preserved
Tree broken

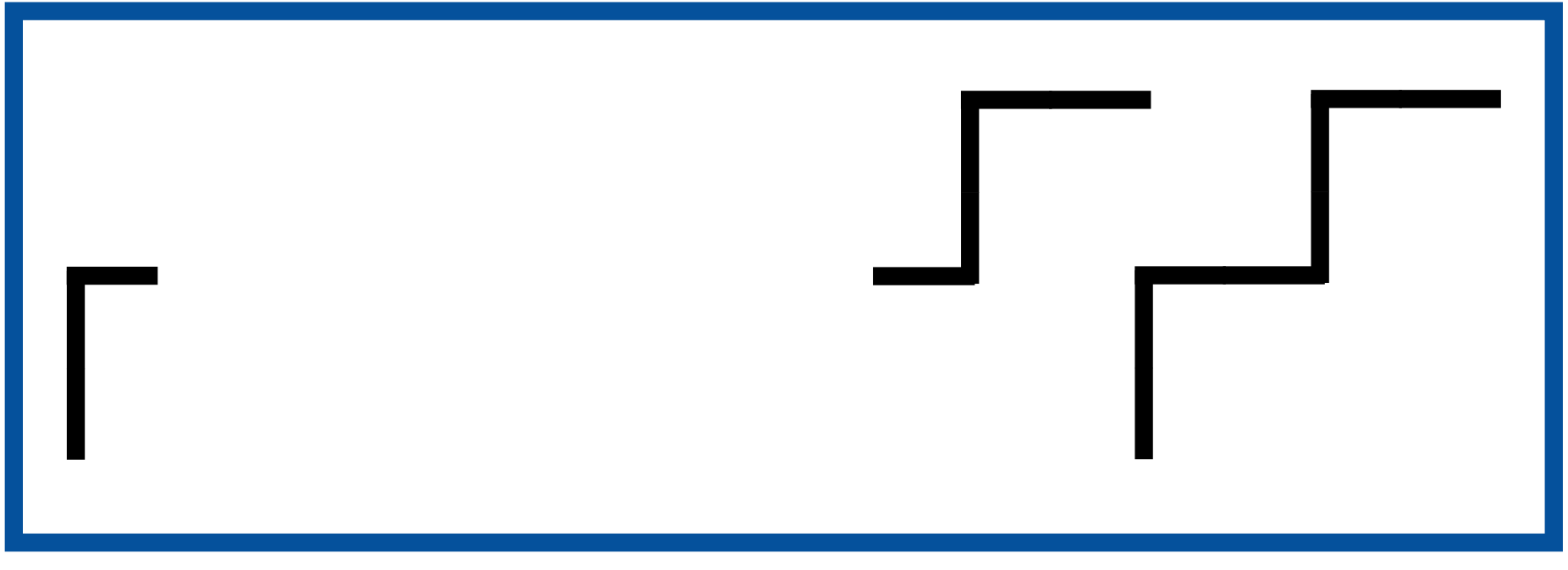

Error rates
Matches
Deviants
***
***
***
0.75
0.50
0.25
Same subtree
Different subtrees
Rotated same
Rotated different

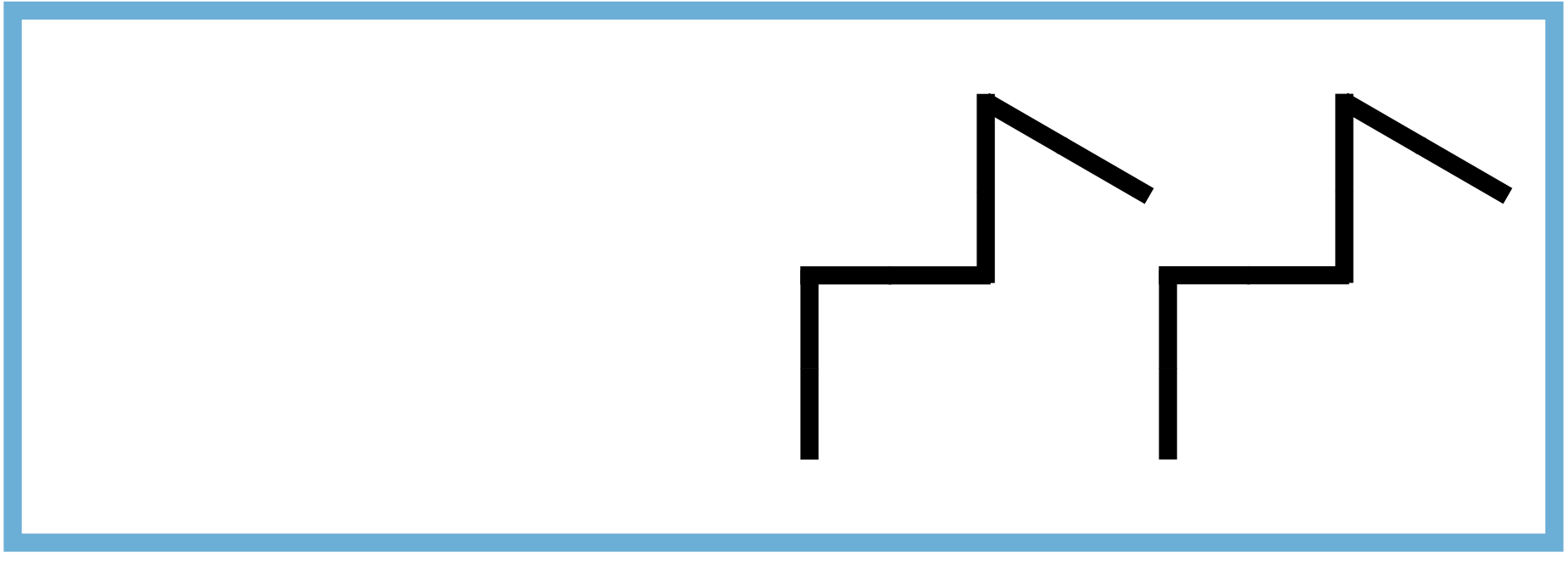


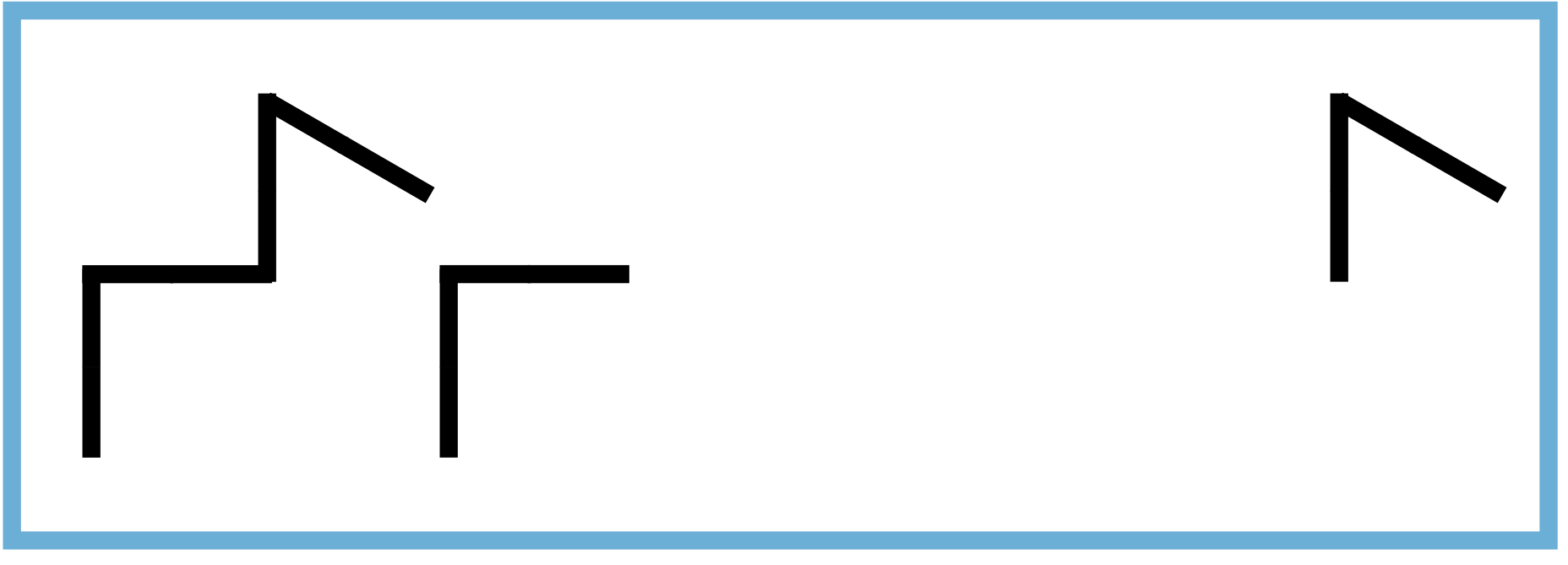


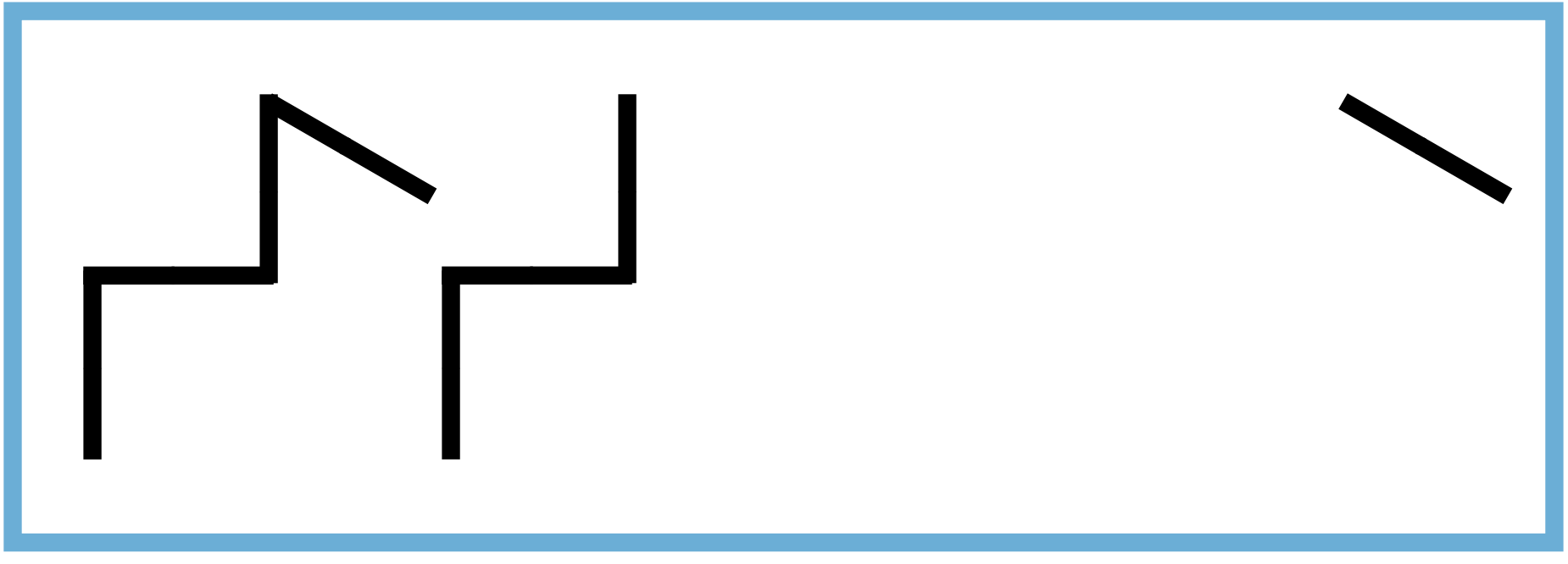


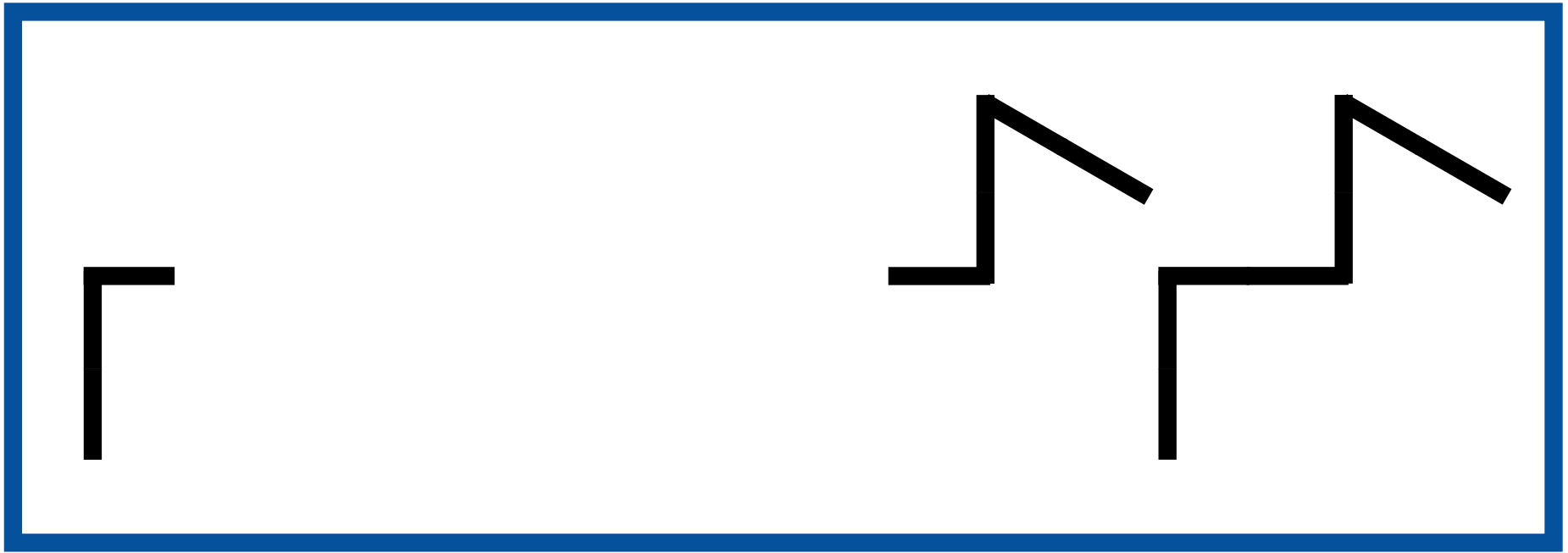


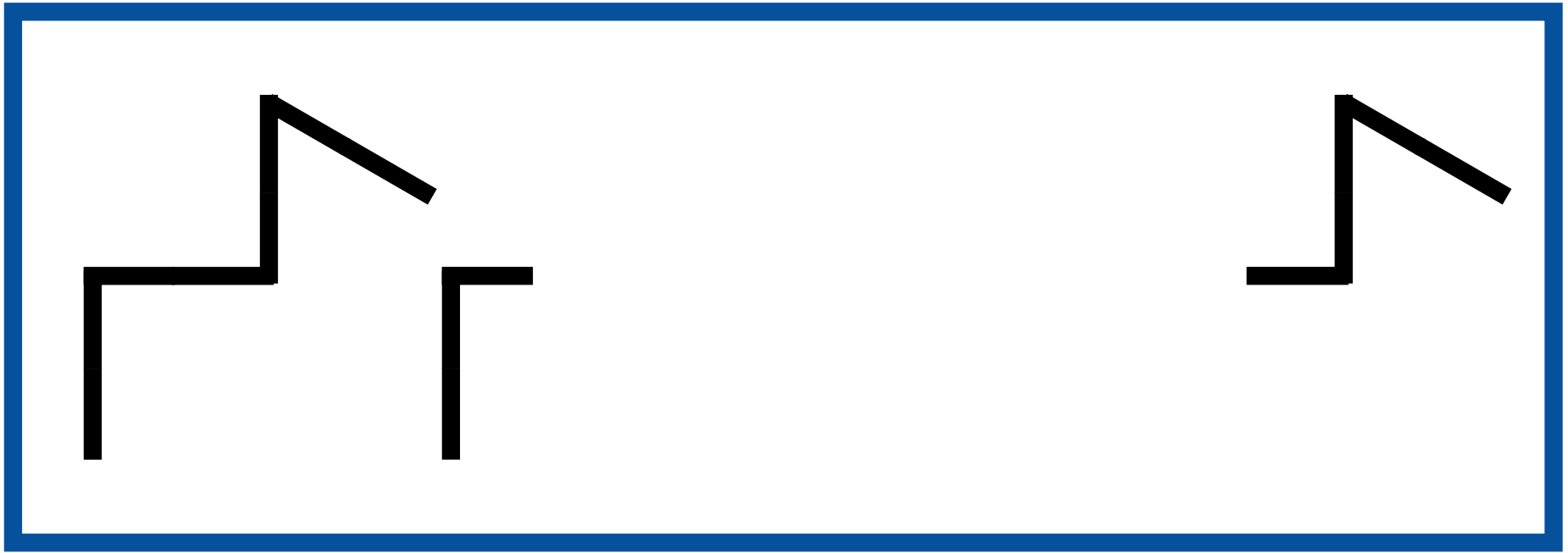


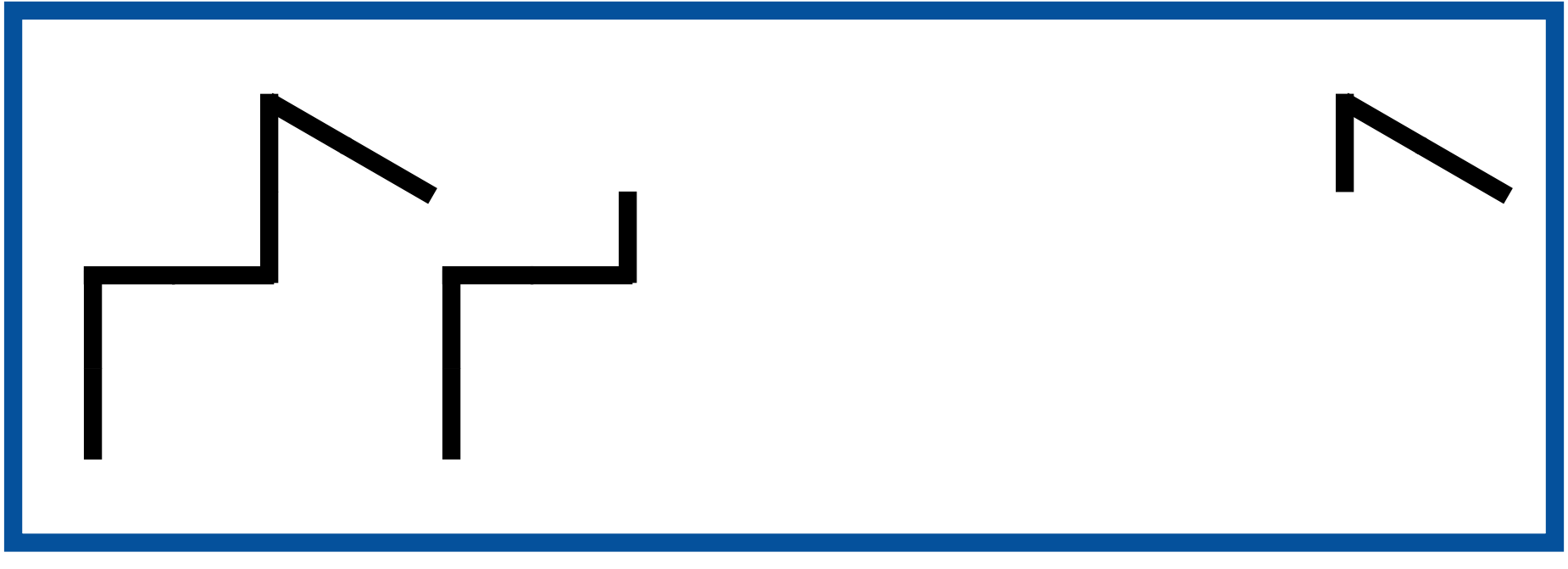


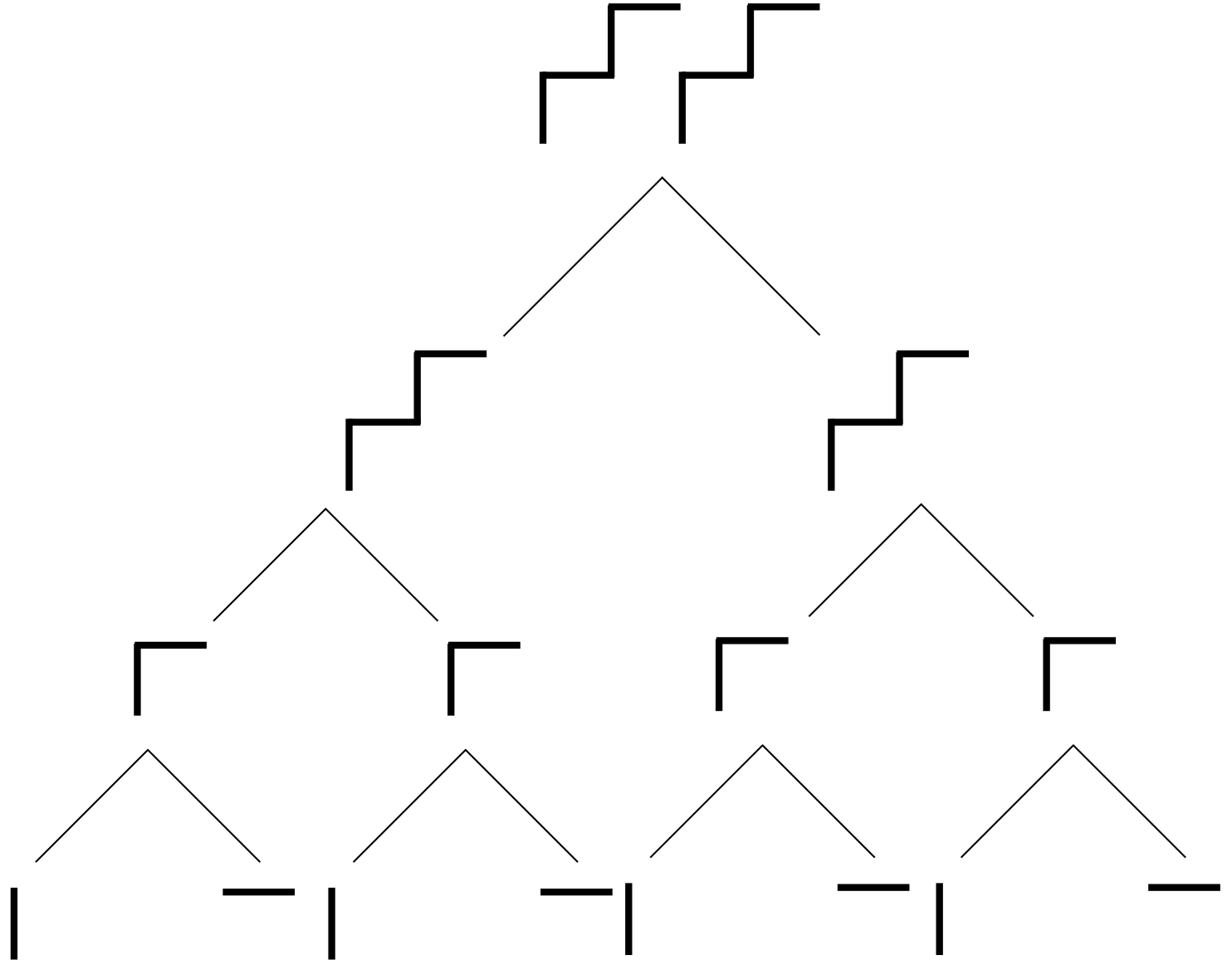


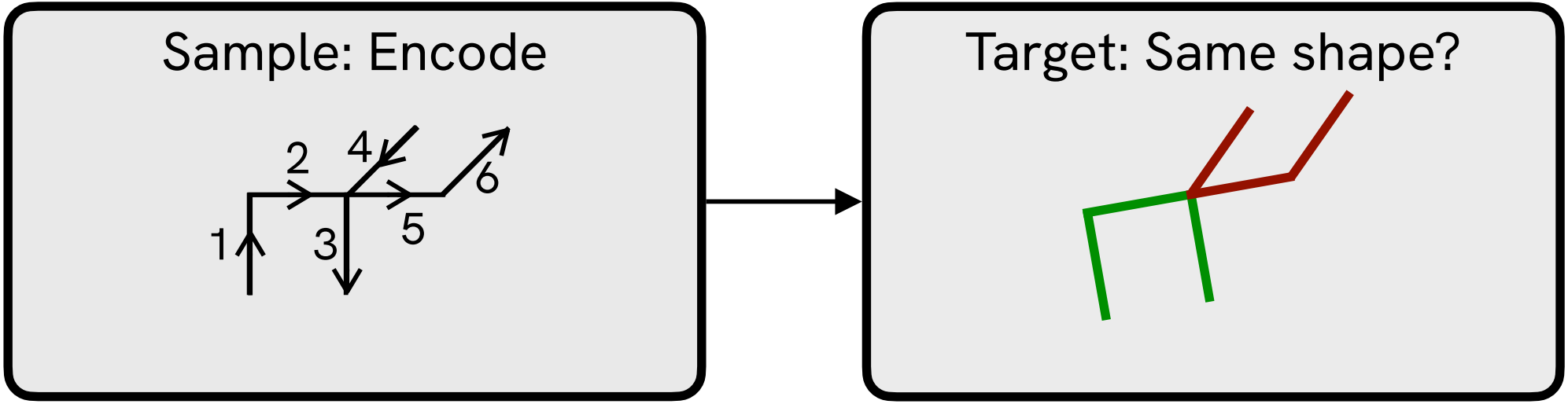


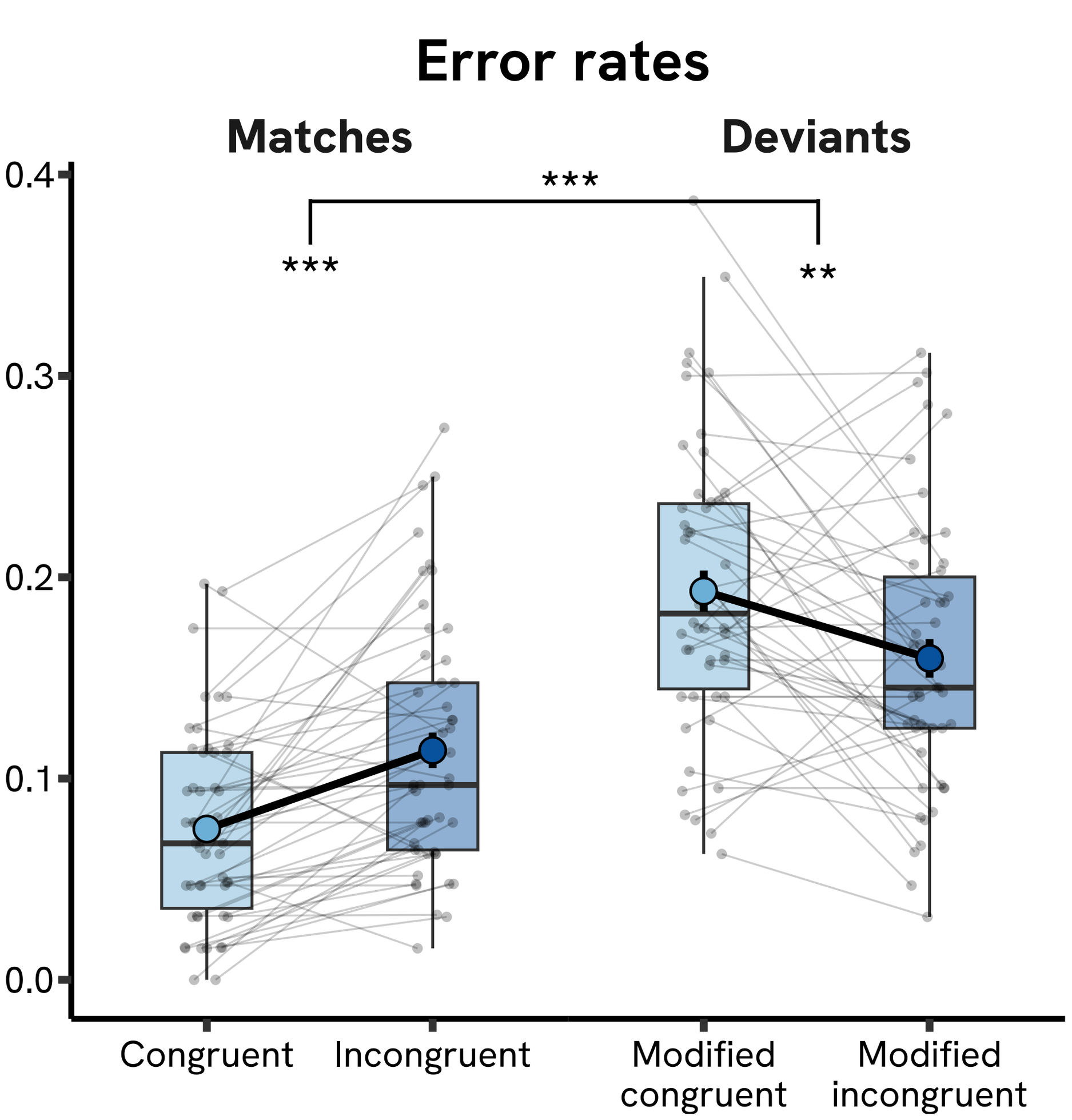


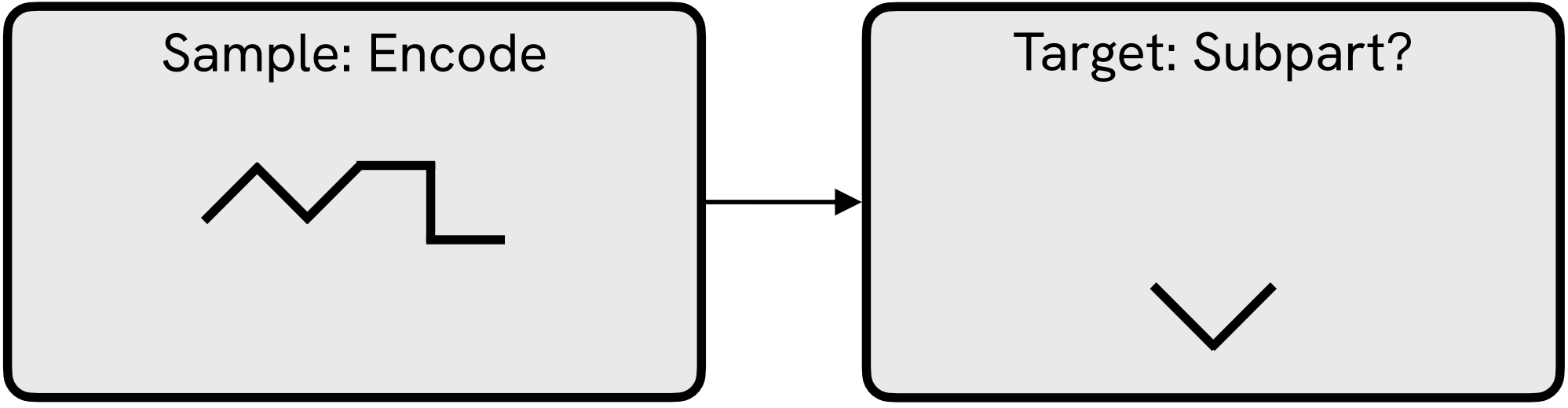


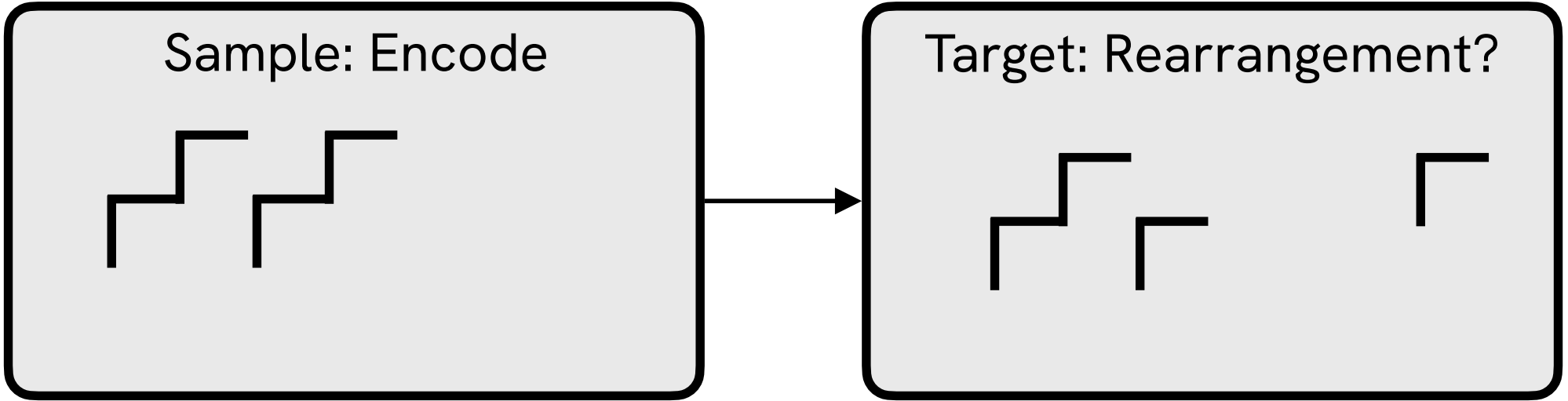





Sample: Encode
1
2
3
4
5
6
Target: Same shape?


Error rates
Matches
Deviants
***
***
**
0.4
0.3
0.2
0.1
0.0
Congruent
Incongruent
Modified congruent
Modified incongruent


Sample: Encode
Target: Subpart?


Sample: Encode
Target: Rearrangement?

Background

We build on the recent proposal that **geometric shapes are represented in a language of thought (LOT)** consisting of a handful of primitives that can be combined to recreate the encoded shape (Sablé-Meyer et al. 2022)

Adults' reaction times and error rates in encoding shapes were predicted by the **minimum description length (MDL)** of the hypothesized LOT

This fuses two aspects of geometric shape representations that are partly independent, **format** (LOT) and **selection** (MDL), and uses MDL data to argue for LOT—highly indirect evidence

Current study provides **more direct evidence for tree structure** in geometric shape representations in 3 online experiments with adults

Geometric shape representations in human adults have syntactic structure

Barbu Revencu and Stanislas Dehaene

NeuroSpin Cognitive Neuroimaging Unit | CEA | INSERM | Université Paris-Saclay

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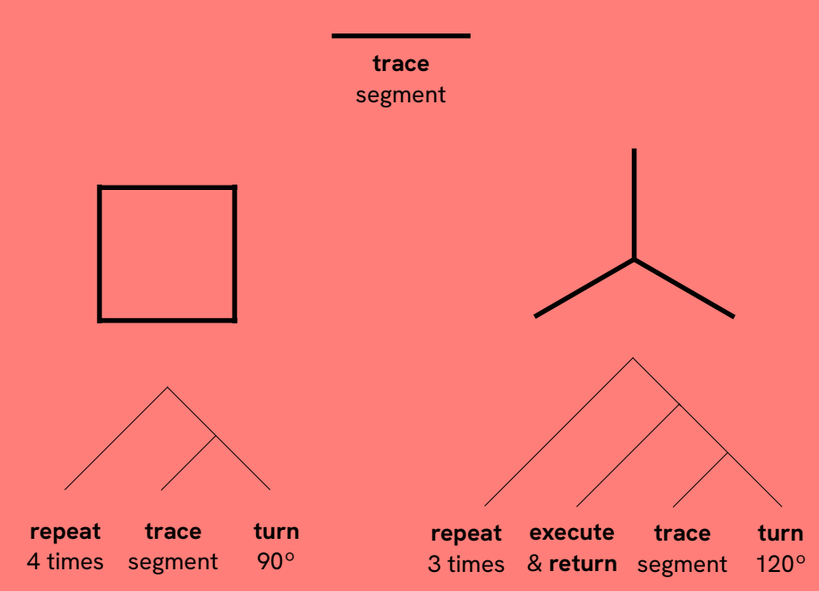
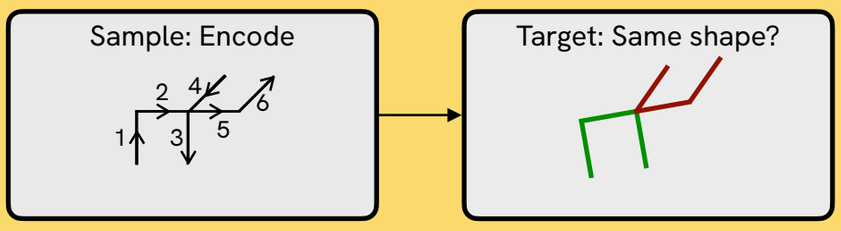
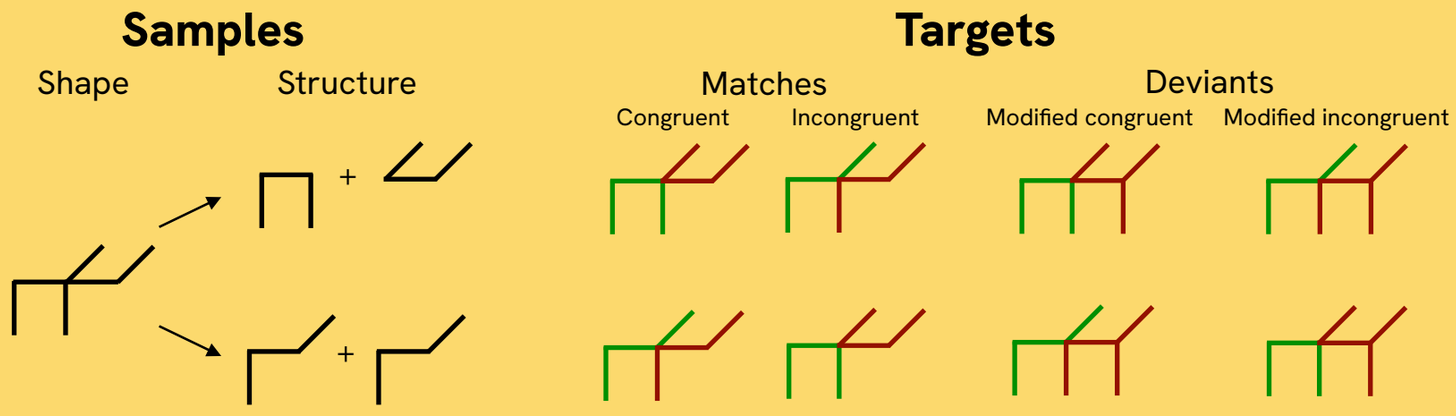
Hypothesis: Geometric shapes are encoded in a dedicated, syntactically structured language of thought



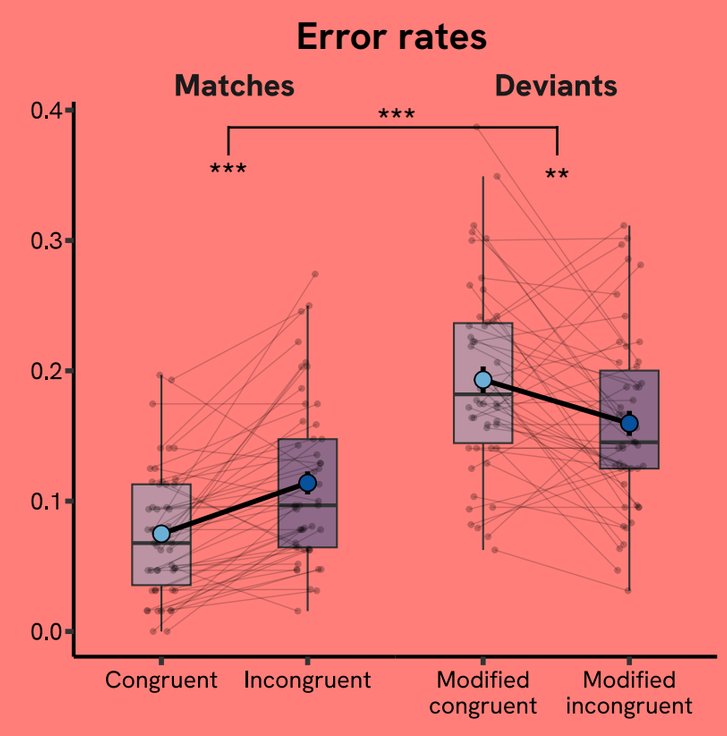
Experiment 1: Structural Ambiguity

50 subjects | 256 trials



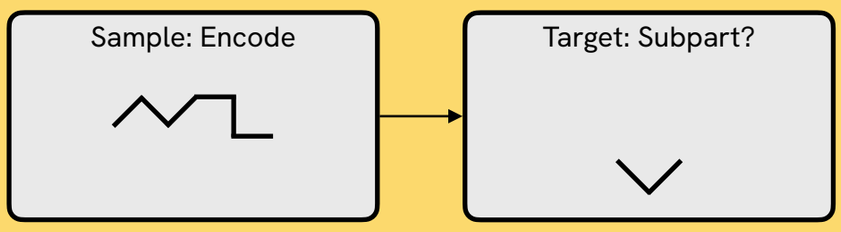


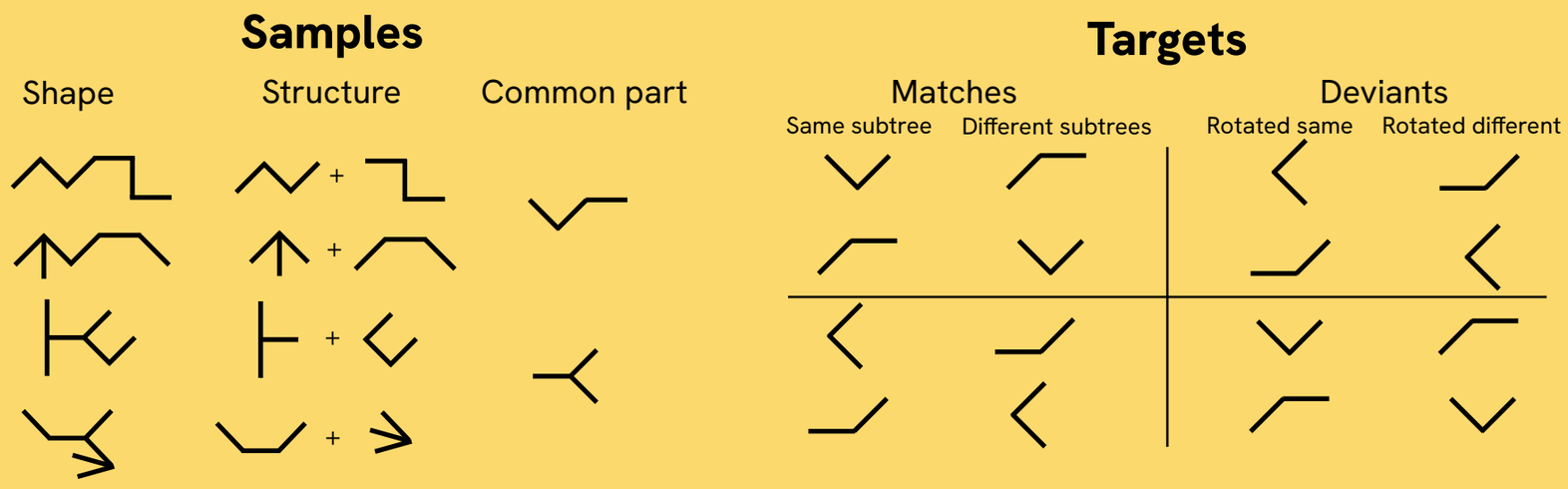
Different structured representations can be induced for the same shape



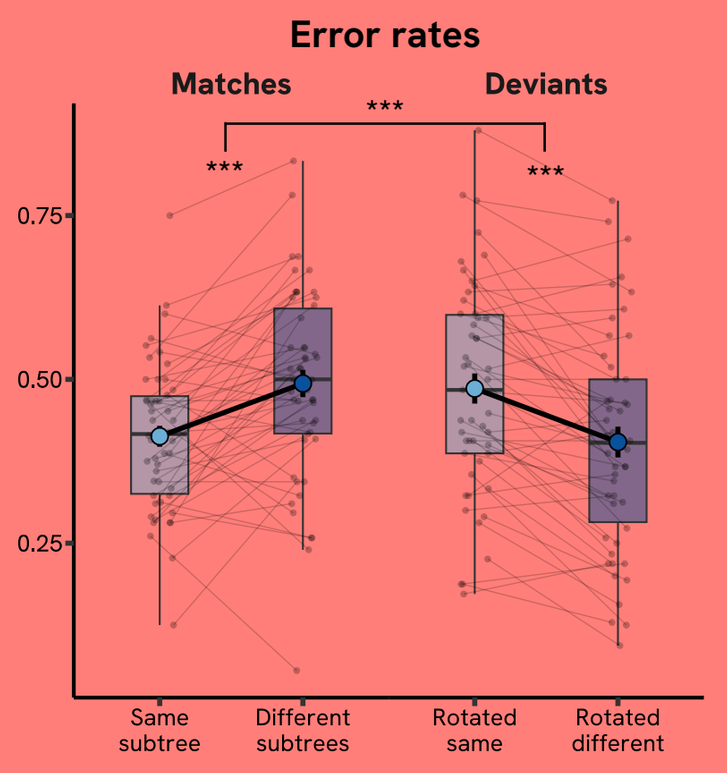
Experiment 2: Subtree Facilitation

50 subjects | 128 trials



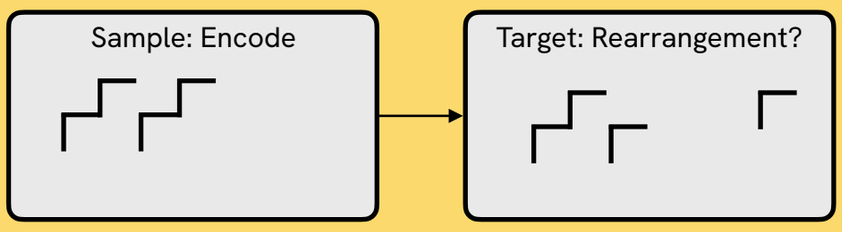


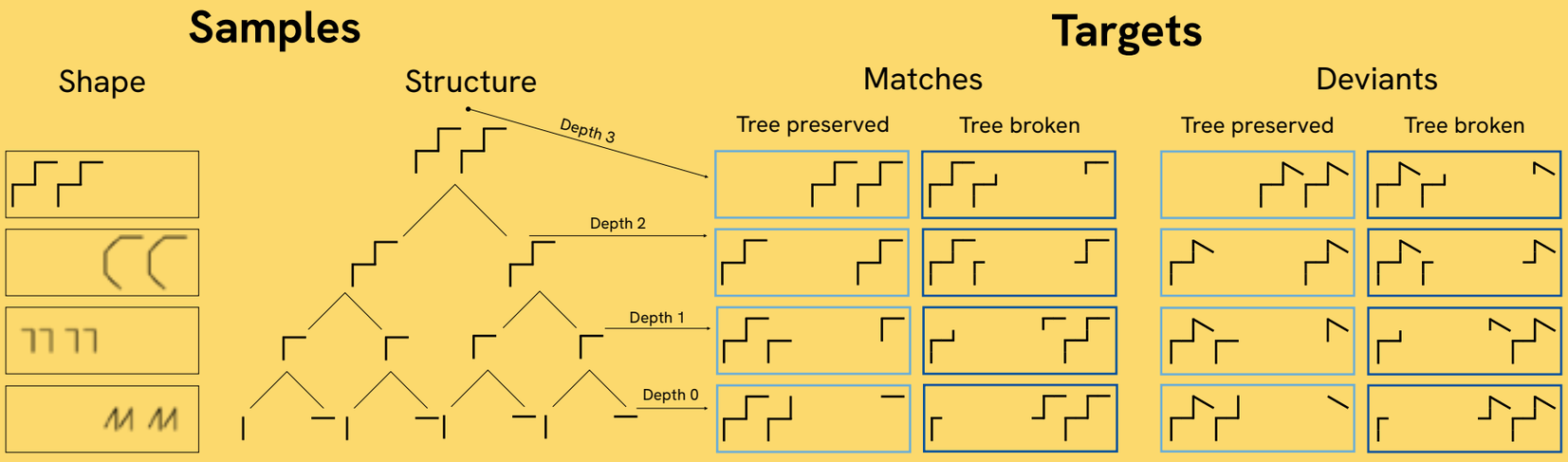
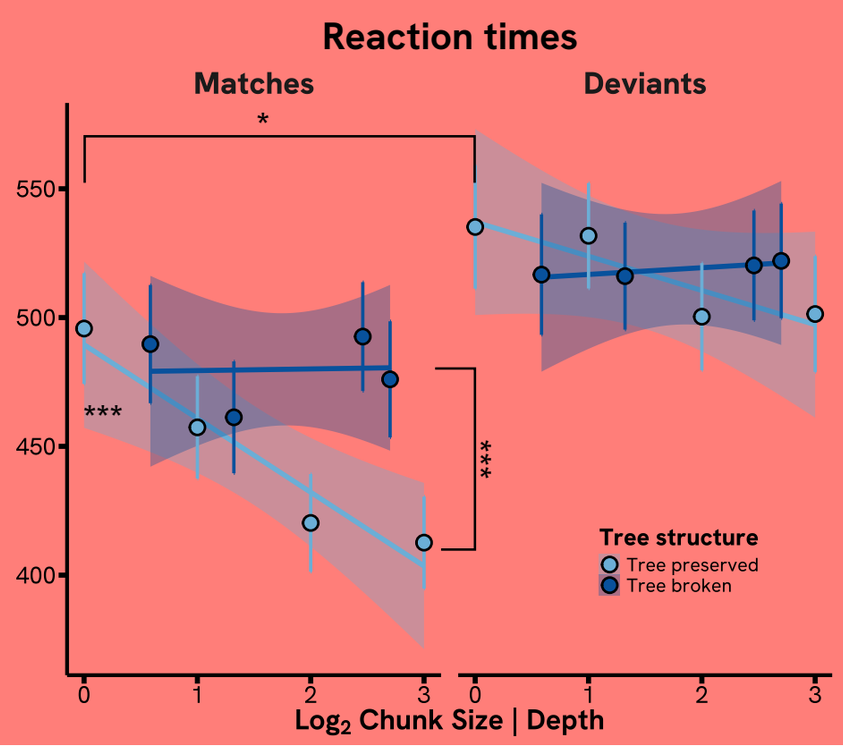
Subparts are easier to recognize when they belong to the same subtree



Experiment 3: Movement Depth

36 subjects | 256 trials





Shapes are easier to reconfigure when they are split higher up in the tree